

Chapter 4

Entity Relationship (ER) Modeling

Database Systems:
Design, Implementation, and
Management, Sixth Edition, Rob and
Coronel

In this chapter, you will learn:

- ❑ How relationships between entities are defined and refined, and how such relationships are incorporated into the database design process
- ❑ How ERD components affect database design and implementation
- ❑ How to interpret the modeling symbols for the four most popular ER modeling tools
- ❑ That real-world database design often requires that you reconcile conflicting goals

The Entity Relationship (ER) Model

- ❑ ER model forms the basis of an ER diagram
- ❑ ERD represents the conceptual database as viewed by end user
- ❑ ERDs depict the ER model's three main components:
 - Entities
 - Attributes
 - Relationships

Entities

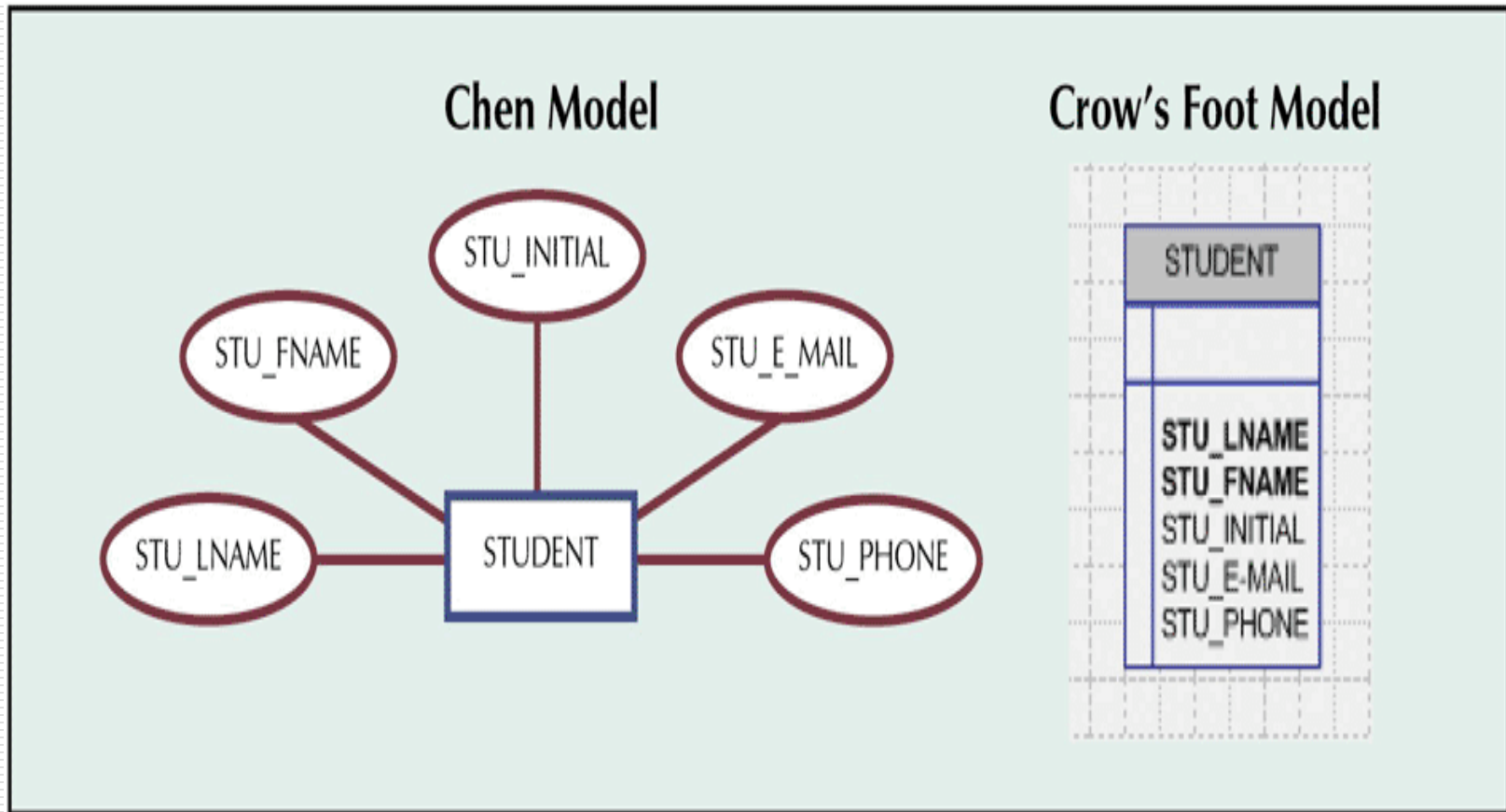
- ❑ Refers to the *entity set* and not to a single entity occurrence
- ❑ Corresponds to a table and not to a row in the relational environment
- ❑ In both the Chen and Crow's Foot models, an entity is represented by a rectangle containing the entity's name
- ❑ Entity name, a noun, is usually written in capital letters

Attributes

- ❑ Characteristics of entities
- ❑ In Chen model, attributes are represented by ovals and are connected to the entity rectangle with a line
- ❑ Each oval contains the name of the attribute it represents
- ❑ In the Crow's Foot model, the attributes are simply written in the attribute box below the entity rectangle

The Attributes of the STUDENT Entity

FIGURE 4.1 THE ATTRIBUTES OF THE STUDENT ENTITY



Domains

- Attributes have a *domain*:
 - The attribute's set of possible values
- Attributes may share a domain
 - For example, the Address attribute for both Customer and Agent can have similar type entries

Primary Keys

- ❑ Underlined in the ER diagram
- ❑ Key attributes are also underlined in a frequently used table structure shorthand
 - CAR(**CAR VIN**,MOD_CODE, CAR_YEAR,CAR_COLOR)
- ❑ Ideally composed of only a single attribute
- ❑ Possible to use a *composite key*:
 - Primary key composed of more than one attribute

The CLASS Table (Entity) Components and Contents

- ❑ The table below can be represented as
 - CLASS(**CLASS_CODE**, CRS_CODE, CLASS_SECTION, CLASS_TIME, CLASS_ROOM, PROF_NUM) or
 - CLASS(**CRS_CODE**, **CLASS_SECTION**, CLASS_CODE, CLASS_TIME, CLASS_ROOM, PROF_NUM)

FIGURE 4.2 THE CLASS TABLE (ENTITY) COMPONENTS AND CONTENTS

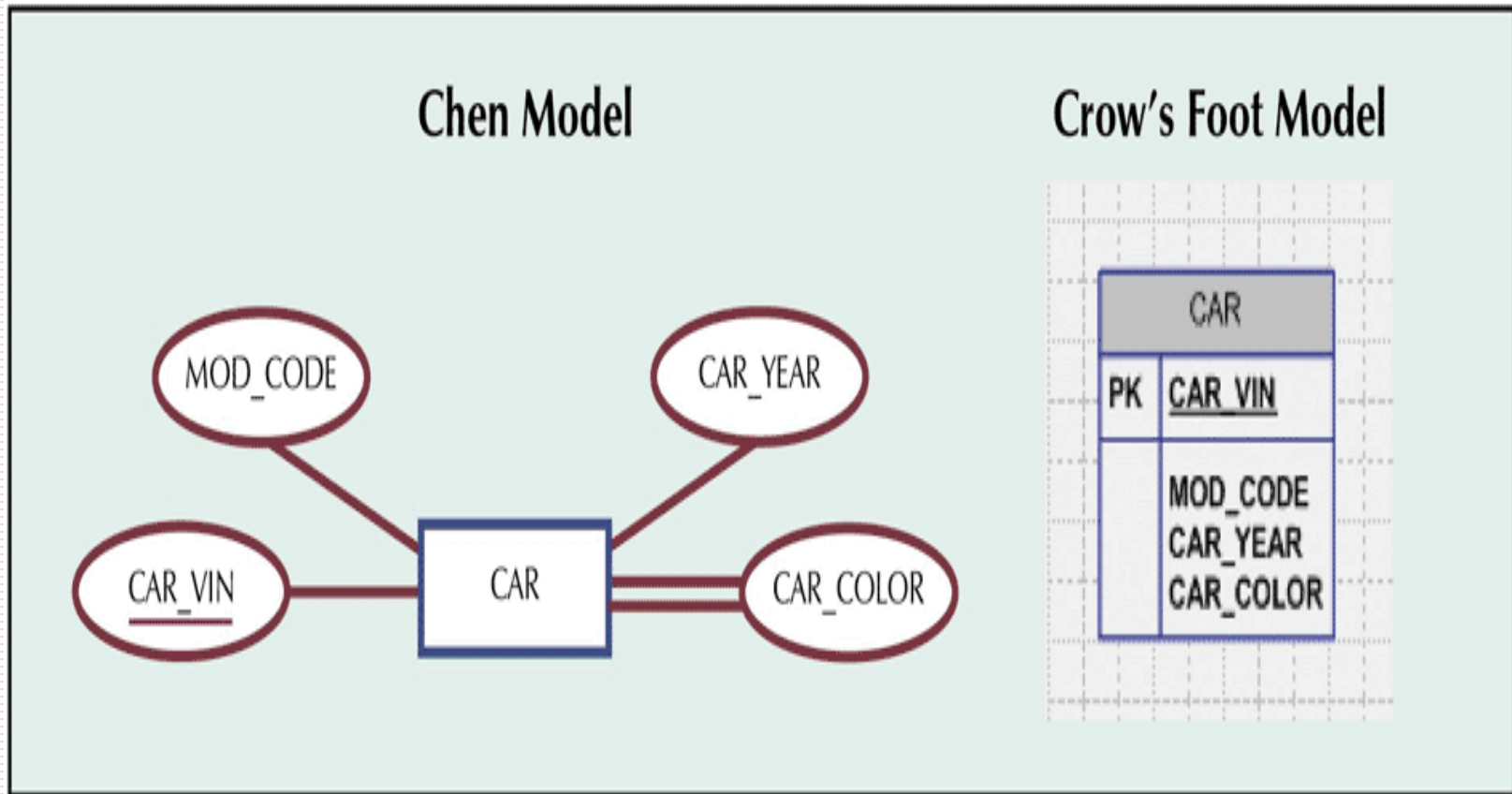
Table name: CLASS		Database name: Ch04_TinyCollege					
		CLASS_CODE	CRS_CODE	CLASS_SECTION	CLASS_TIME	CLASS_ROOM	PROF_NUM
▶	+	10012	ACCT-211	1	MWF 8:00-8:50 a.m.	BUS311	105
	+	10013	ACCT-211	2	MWF 9:00-9:50 a.m.	BUS200	105
	+	10014	ACCT-211	3	TTh 2:30-3:45 p.m.	BUS252	342
	+	10015	ACCT-212	1	MWF 10:00-10:50 a.m.	BUS311	301
	+	10016	ACCT-212	2	Th 6:00-8:40 p.m.	BUS252	301
	+	10017	CIS-220	1	MWF 9:00-9:50 a.m.	KLR209	228
	+	10018	CIS-220	2	MWF 9:00-9:50 a.m.	KLR211	114
	+	10019	CIS-220	3	MWF 10:00-10:50 a.m.	KLR209	228
	+	10020	CIS-420	1	W 6:00-8:40 p.m.	KLR209	162
	+	10021	QM-261	1	MWF 8:00-8:50 a.m.	KLR200	114
	+	10022	QM-261	2	TTh 1:00-2:15 p.m.	KLR200	114
	+	10023	QM-362	1	MWF 11:00-11:50 a.m.	KLR200	162
	+	10024	QM-362	2	TTh 2:30-3:45 p.m.	KLR200	162

Attributes

- ❑ Composite attribute
 - Not to be confused with composite key.
 - This is an attribute that can be broken down into more atomic attributes
 - ❑ Address can be divided into street, city, state and zip
- ❑ Simple attribute– no further division possible
- ❑ Single-value attribute – can have only one value (social security number)
- ❑ Multivalued attributes – a household may have several phone numbers
 - Denoted with a double connecting line in the Chen model

A Multivalued Attribute in an Entity

FIGURE 4.3 A MULTIVALUED ATTRIBUTE IN AN ENTITY

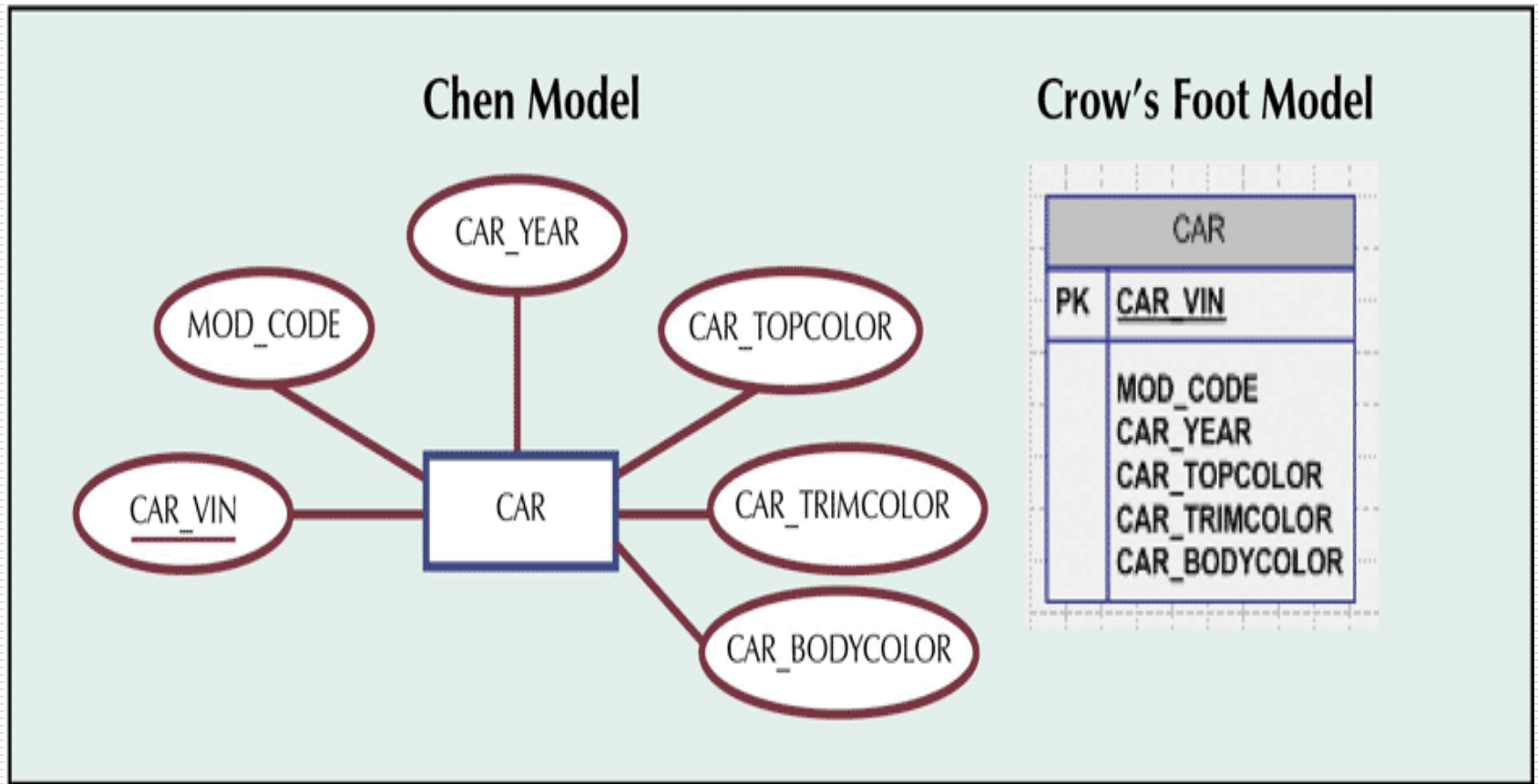


Resolving Multivalued Attribute Problems

- ❑ Although the conceptual model can handle multivalued attributes, *you should not implement them in the relational DBMS*. Instead, follow one of these two options
 1. Within original entity, create several new attributes, one for each of the original multivalued attribute's components
 - CAR_COLOR can be split into CAR_TOPCOLOR, CAR_BODYCOLOR and CAR_TRIMCOLOR
 - Can lead to major structural problems in the table.
 - If some cars have many types of colors and others have few colors, then all cars need to have attributes to handle the maximum number of colors. But many of those fields will be null for many rows.

Splitting the Multivalued Attribute into New Attributes

FIGURE 4.4 SPLITTING THE MULTIVALUED ATTRIBUTE INTO NEW ATTRIBUTES



Components of the Multivalued Attribute

TABLE 4.1 COMPONENTS OF THE MULTIVALUED ATTRIBUTE

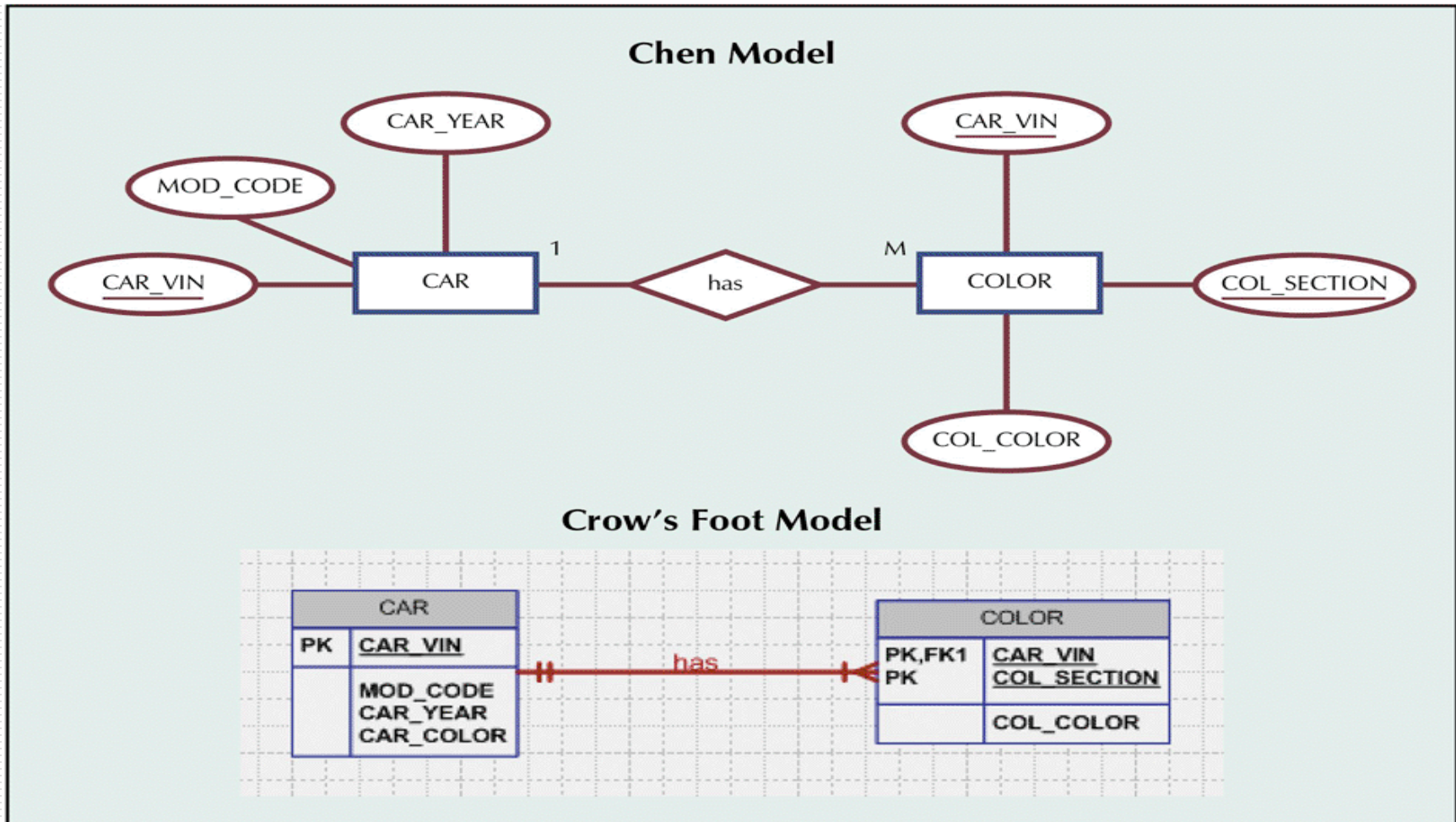
SECTION	COLOR
Top	White
Body	Blue
Trim	Gold
Interior	Blue

Resolving Multivalued Attribute Problems

2. Create a new entity composed of the original multivalued attribute's components.
 - The new entity is related to the original entity in a 1:M relationship
 - Color needs to be defined only for those sections that have color. This is done in the COL_SECTION attribute

A New Entity Set Composed of a Multivalued Attribute's Components

FIGURE 4.5 A NEW ENTITY SET COMPOSED OF A MULTIVALUED ATTRIBUTE'S COMPONENTS

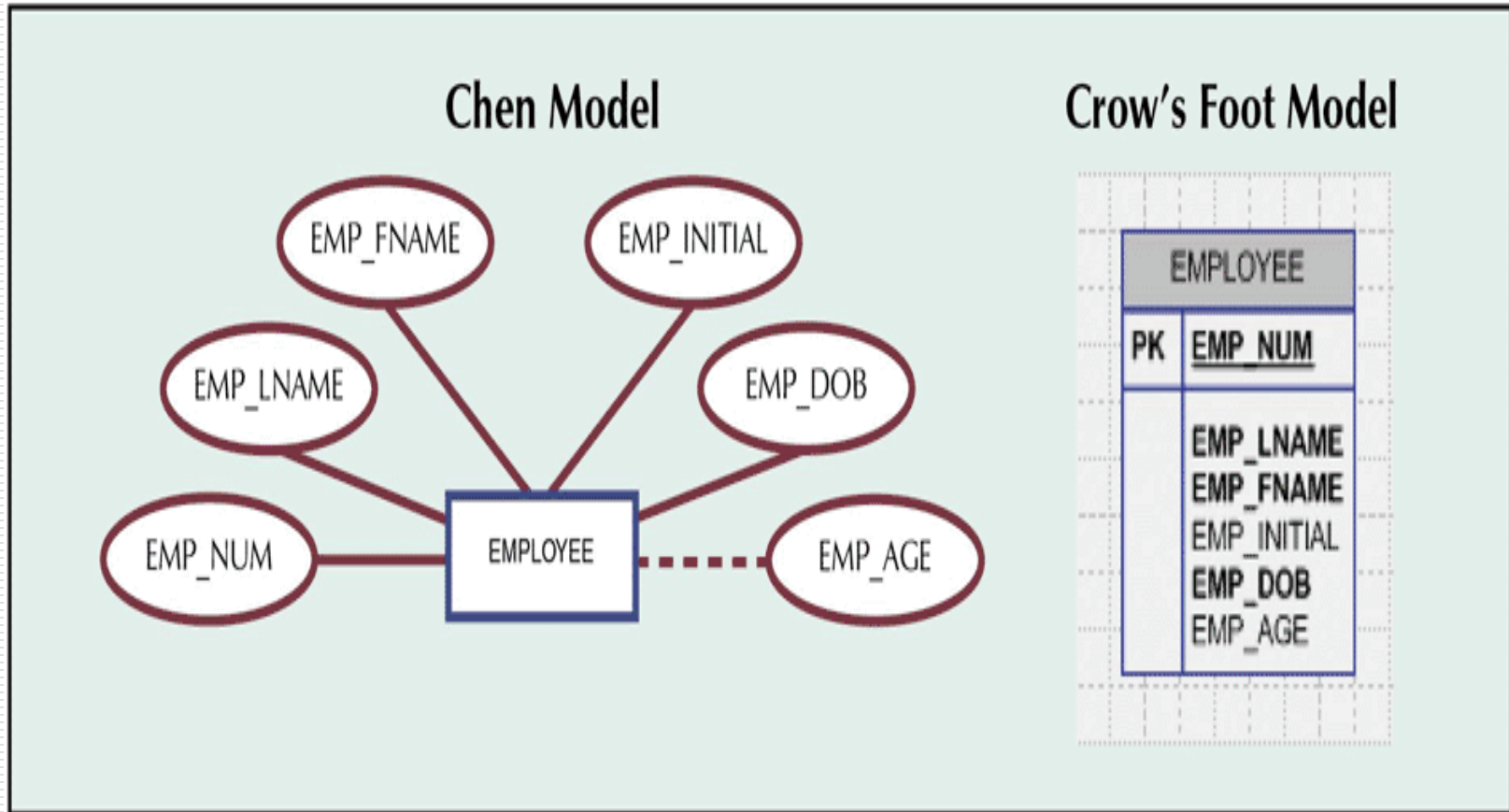


Derived Attributes

- ❑ Attribute whose value may be calculated (derived) from other attributes
 - Age can be calculated by subtracting date of birth from current date
- ❑ Need not be physically stored within the database but can be based on processing requirements
- ❑ Can be derived by using an algorithm
- ❑ Denoted by a dashed line in the Chen model

Depiction of a Derived Attribute

FIGURE 4.6 DEPICTION OF A DERIVED ATTRIBUTE



Relationships

- ❑ Association between entities
- ❑ Participants:
 - Entities that participate in a relationship
- ❑ Relationships between entities always operate in both directions
- ❑ Relationship classification is difficult to establish if you only know one side
 - A DIVISION is managed by one EMPLOYEE
 - ❑ Don't know if this is 1:1 or 1:M, must know if an EMPLOYEE can manage more than one DIVISION

Connectivity and Cardinality

□ Connectivity

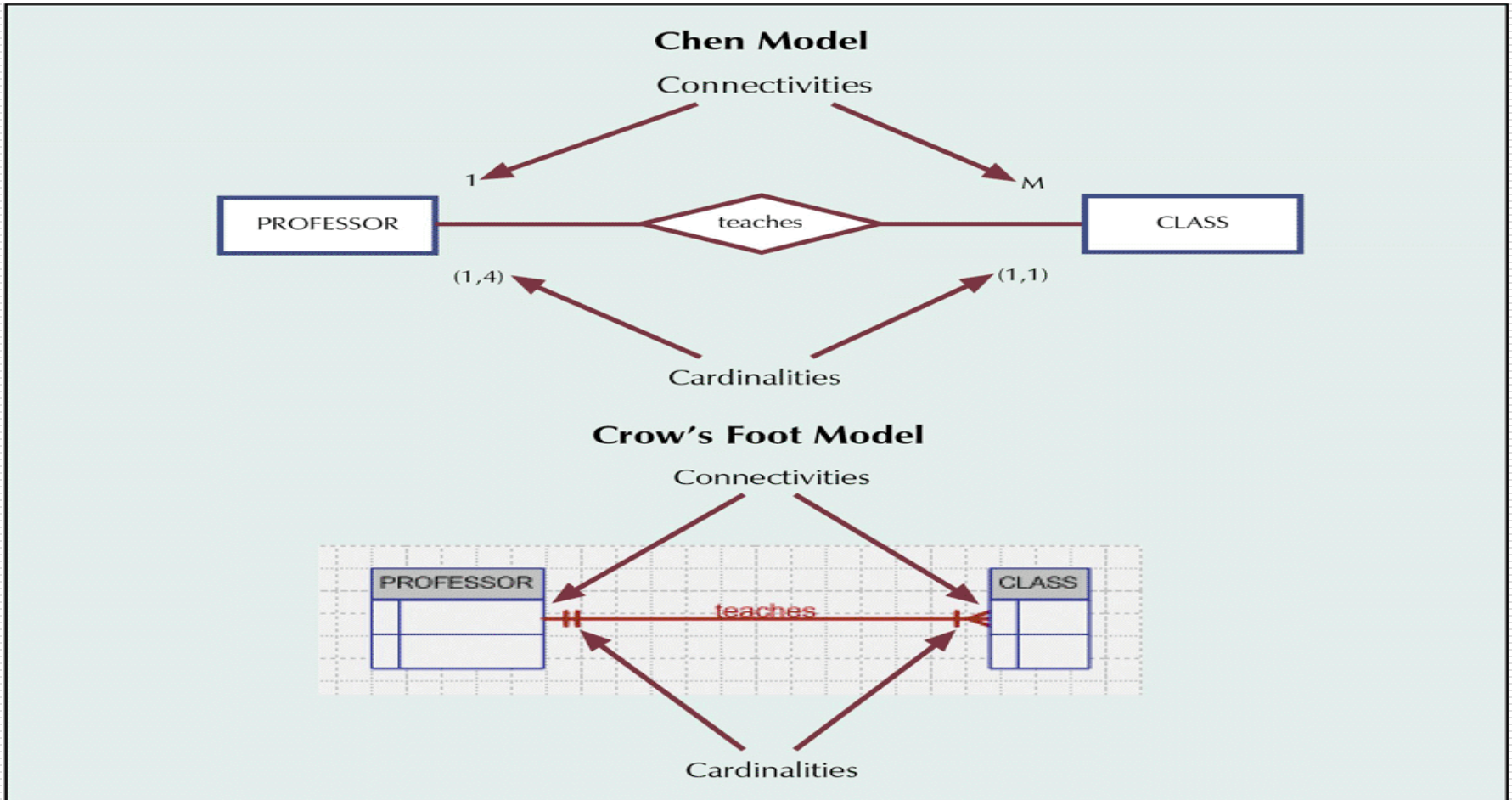
- Used to describe the relationship classification

□ Cardinality

- Expresses the specific number of entity occurrences associated with one occurrence of the related entity
- Established by very concise statements known as *business rules*

Connectivity and Cardinality in an ERD

FIGURE 4.7 CONNECTIVITY AND CARDINALITY IN AN ERD



RELATIONSHIP Strength

- ☐ Existence dependence
 - Entity's existence depends on the existence of one or more other entities
 - ☐ EMPLOYEE claims DEPENDENT
- ☐ Existence independence
 - Entity can exist apart from one or more related entities
 - ☐ PART supplied by VENDOR (some parts may be in-house)

RELATIONSHIP Strength

- Weak (non-identifying) relationships
 - One entity is not existence-independent on another entity
 - The PK of the related entity does not contain a PK component of the parent entity. The CLASS PK did not inherit the PK component from the COURSE entit

COURSE(**CRS_CODE**, DEPT_CODE, CRS_DESC, CRS_CREDIT)
CLASS(**CLASS_CODE**, CRS-CODE, CLASS_SECTION,...)

A Weak (Non-Identifying) Relationship Between COURSE and CLASS

FIGURE 4.8 A WEAK (NON-IDENTIFYING) RELATIONSHIP BETWEEN COURSE AND CLASS



FIGURE 4.9 A WEAK RELATIONSHIP BETWEEN COURSE AND CLASS

Table name: COURSE

Database name: Ch04_TinyCollege

	CRS_CODE	DEPT_CODE	CRS_DESCRIPTION	CRS_CREDIT
▶	ACCT-211	ACCT	Accounting I	3
+	ACCT-212	ACCT	Accounting II	3
+	CIS-220	CIS	Intro. to Microcomputing	3
+	CIS-420	CIS	Database Design and Implementation	4
+	MATH-243	MATH	Mathematics for Managers	3
+	QM-261	CIS	Intro. to Statistics	3
+	QM-362	CIS	Statistical Applications	4

Table name: CLASS

	CLASS_CODE	CRS_CODE	CLASS_SECTION	CLASS_TIME	ROOM_CODE	PROF_NUM
▶	10012	ACCT-211	1	MVVF 8:00-8:50 a.m.	BUS311	105
+	10013	ACCT-211	2	MVVF 9:00-9:50 a.m.	BUS200	105
+	10014	ACCT-211	3	TTh 2:30-3:45 p.m.	BUS252	342
+	10015	ACCT-212	1	MVVF 10:00-10:50 a.m.	BUS311	301
+	10016	ACCT-212	2	Th 6:00-8:40 p.m.	BUS252	301
+	10017	CIS-220	1	MVVF 9:00-9:50 a.m.	KLR209	228
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+	10020	CIS-420	1	vV 6:00-8:40 p.m.	KLR209	162
+	10021	QM-261	1	MVVF 8:00-8:50 a.m.	KLR200	114
+	10022	QM-261	2	TTh 1:00-2:15 p.m.	KLR200	114
+	10023	QM-362	1	MVVF 11:00-11:50 a.m.	KLR200	162
+	10024	QM-362	2	TTh 2:30-3:45 p.m.	KLR200	162
+	10025	MATH-243	1	Th 6:00-8:40 p.m.	DRE155	325

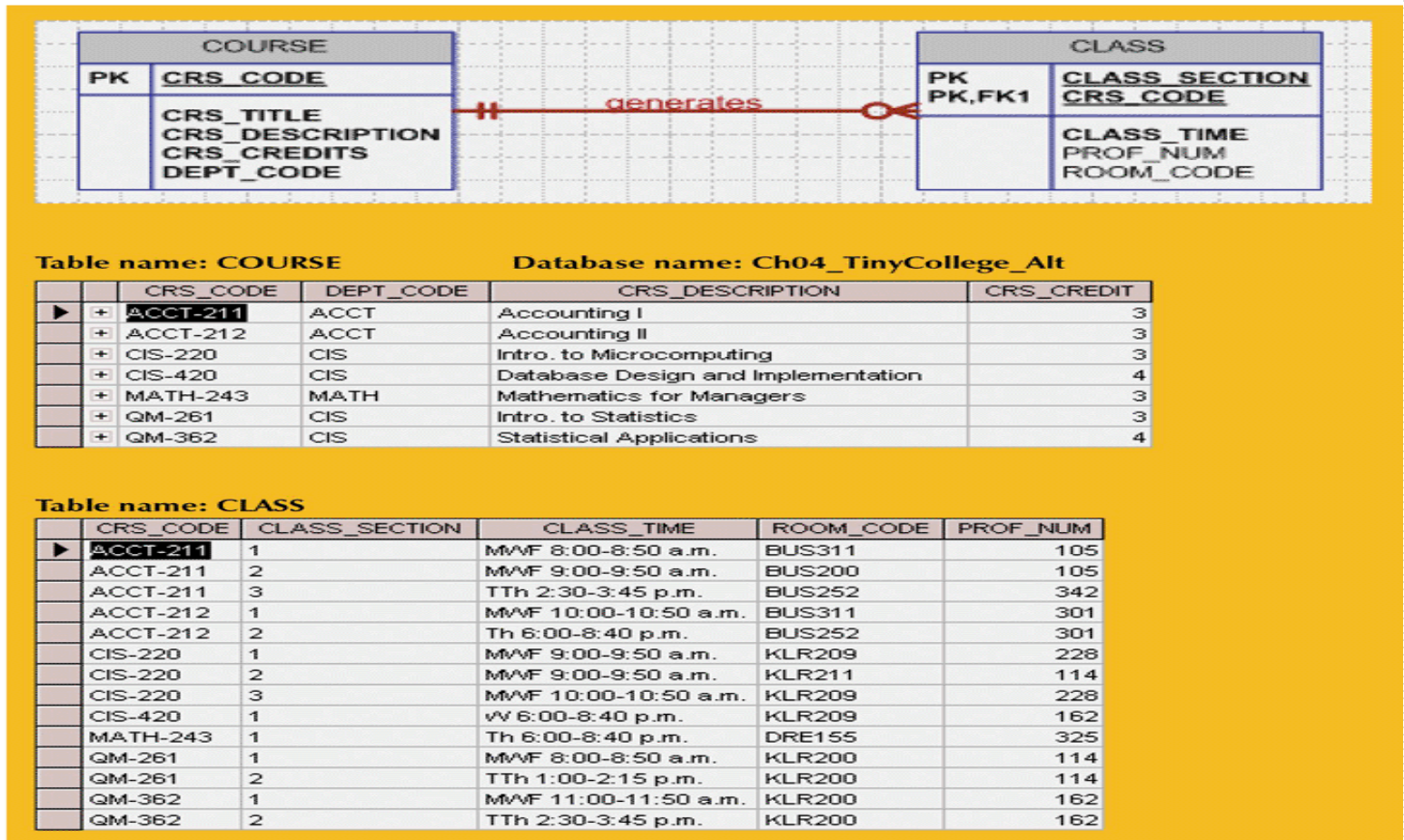
RELATIONSHIP Strength

- Strong (Identifying) Relationships
 - Related entities are existence-dependent
 - Whenever the PK of the related entity contains a PK component of the parent entity

COURSE(**CRS_CODE**, DEPT_CODE, CRS_DESC, CRS_CREDIT)
CLASS(**CRS_CODE**, **CLASS_CODE**, CRS-CODE,
CLASS_SECTION,...)

A Strong (Identifying) Relationship Between COURSE and CLASS

FIGURE 4.10 A STRONG (IDENTIFYING) RELATIONSHIP BETWEEN COURSE AND CLASS



Relationship Participation

☐ Optional:

- One entity occurrence does not *require* a corresponding entity occurrence in a particular relationship
 - ☐ COURSE generates CLASS – some courses may not generate a class
 - ☐ A small circle is drawn on the side of the optional entity in the Chen and Crow's foot models)

☐ Mandatory:

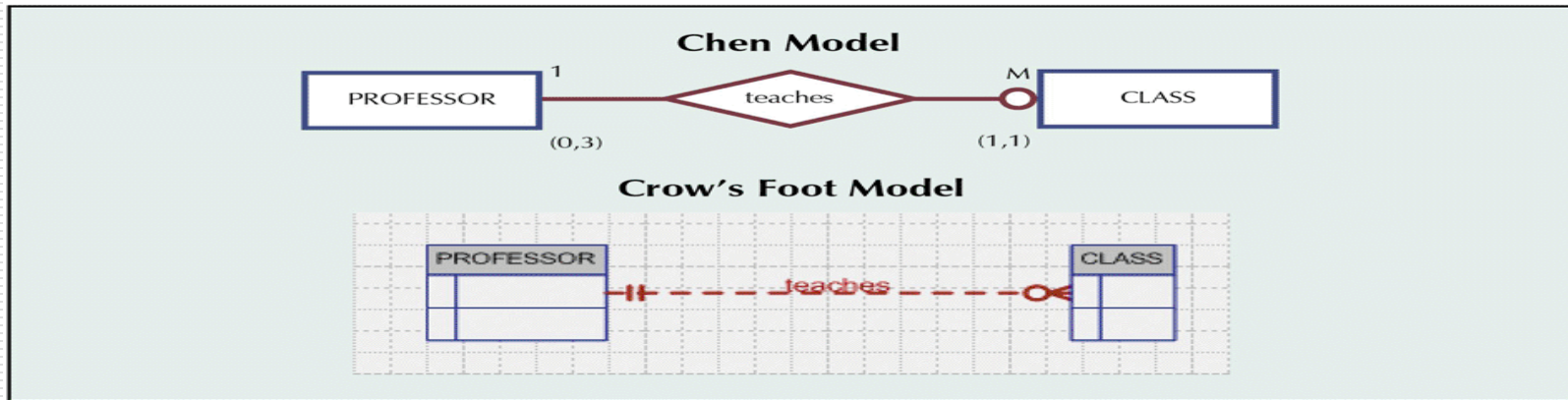
- One entity occurrence *requires* a corresponding entity occurrence in a particular relationship
 - ☐ Minimum cardinality of 1

Relationship Participation and Strength

- ❑ Incorrect to conclude that relationships are weak when they are created between optional entities and strong between mandatory entities
- ❑ You can have a strong relationship with optional relationship participation
 - EMPLOYEE and DEPENDENT – strong relationship but an employee may have no dependents (more on this soon)

An Optional CLASS Entity in the Relationship PROFESSOR teaches CLASS

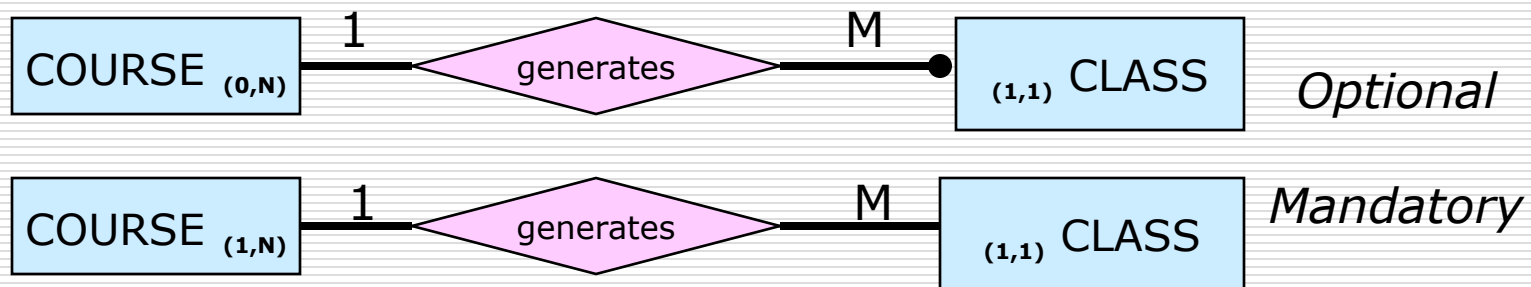
FIGURE 4.11 AN OPTIONAL CLASS ENTITY IN THE RELATIONSHIP PROFESSOR TEACHES CLASS



- ❑ A PROFESSOR may not teach a CLASS – CLASS is optional to PROFESSOR
- ❑ A CLASS must be taught by a PROFESSOR – PROFESSOR is mandatory to CLASS
 - (0,3) means a PROFESSOR can teach no courses and up to a maximum of 3
 - (1,1) means a CLASS has exactly one PROFESSOR

COURSE and CLASS

- What does the relationship “COURSE generates CLASS” imply
 - CLASS is optional – there may be courses with no classes (not offered each semester)
 - CLASS is mandatory – each COURSE must have a least one COURSE

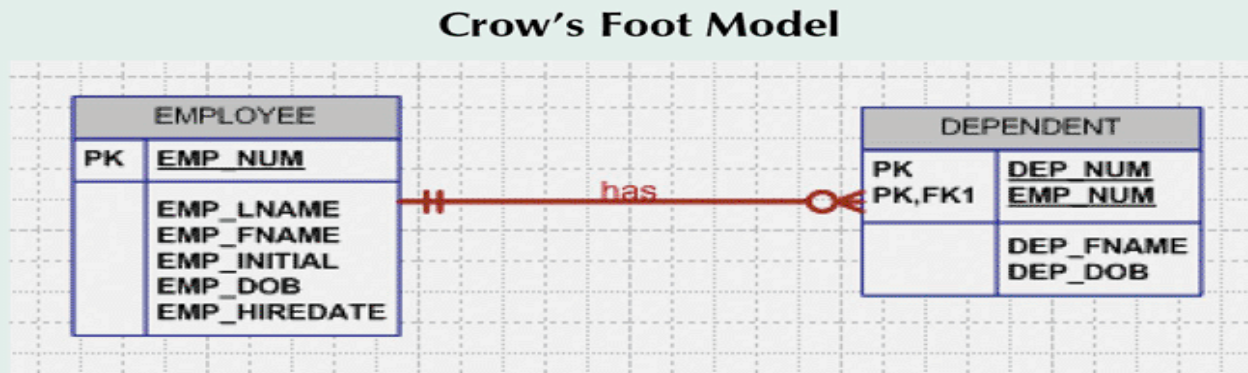
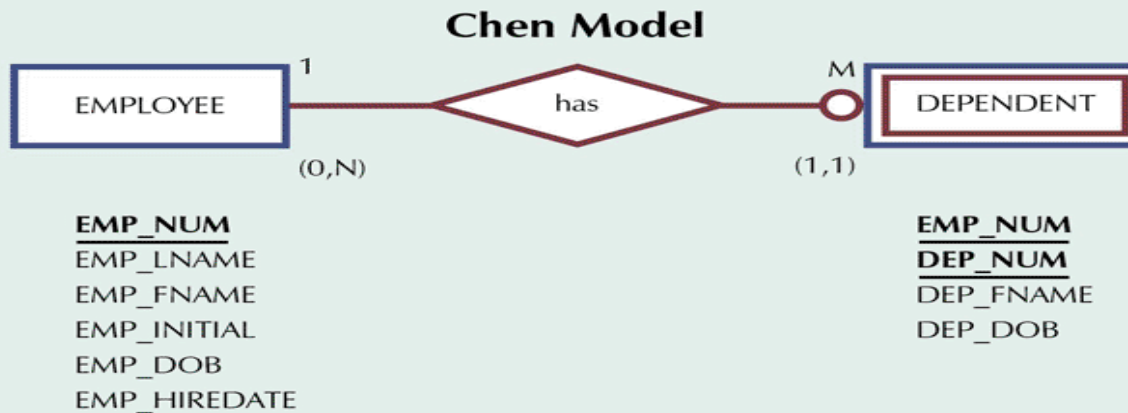


Relationship Strength and Weak Entities

- ❑ Weak entity meets two conditions
 - Existence-dependent:
 - ❑ Cannot exist without entity with which it has a relationship
 - Has primary key that is partially or totally derived from the parent entity in the relationship
 - EMPLOYEE has DEPENDENT (see next slide)
 - In the Chen mode, the weak entity has a double border
 - In the Crow's foot model, the PK/FK designation is used
- ❑ Database designer usually determines whether an entity can be described as weak based on the business rules

A Weak Entity in an ERD

FIGURE 4.14 A WEAK ENTITY IN AN ERD



A Weak Entity in a Strong Relationship

FIGURE 4.15 A WEAK ENTITY IN A STRONG RELATIONSHIP

Table name: EMPLOYEE

Database name: Ch04_ShortCo

		EMP_NUM	EMP_LNAME	EMP_FNAME	EMP_INITIAL	EMP_DOB	EMP_HIREDATE
►	+	1001	Callifante	Jeanine	J	12-Mar-64	25-May-97
	+	1002	Smithson	William	K	23-Nov-70	28-May-97
	+	1003	Washington	Herman	H	15-Aug-68	28-May-97
	+	1004	Chen	Lydia	B	23-Mar-74	15-Oct-98
	+	1005	Johnson	Melanie		28-Sep-66	20-Dec-98
	+	1006	Ortega	Jorge	G	12-Jul-79	05-Jan-02
	+	1007	O'Donnell	Peter	D	10-Jun-71	23-Jun-02
	+	1008	Brzenski	Barbara	A	12-Feb-70	01-Nov-03

Table name: DEPENDENT

	EMP_NUM	DEP_NUM	DEP_FNAME	DEP_DOB
►	1001	1	Annelise	05-Dec-97
	1001	2	Jorge	30-Sep-02
	1003	1	Suzanne	25-Jan-04
	1006	1	Carlos	25-May-01
	1008	1	Michael	19-Feb-95
	1008	2	George	27-Jun-98
	1008	3	Katherine	18-Aug-03

Relationship Degree

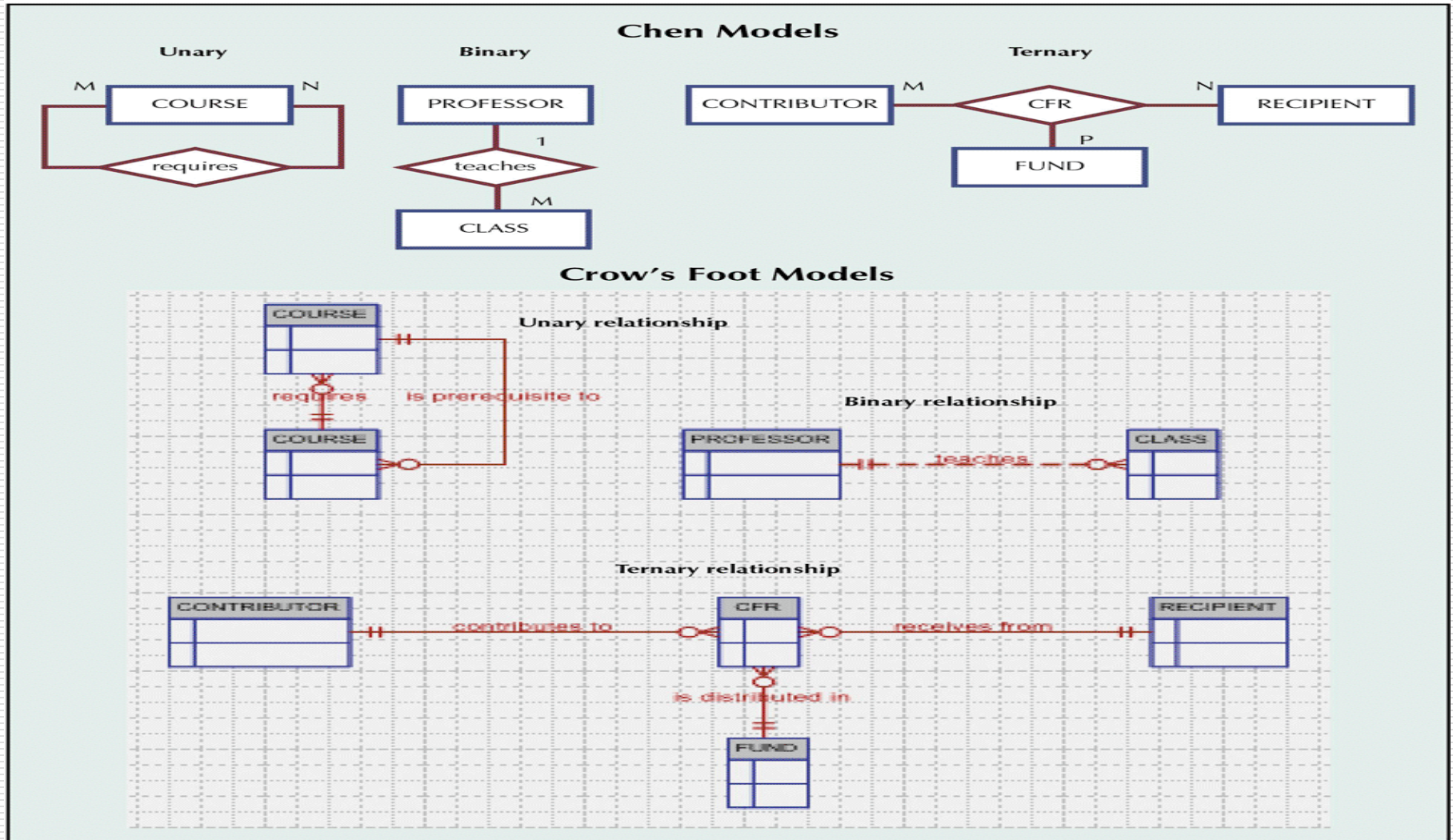
- ❑ Indicates number of associated entities or participants
- ❑ Unary relationship
 - Association is maintained within a single entity
- ❑ Binary relationship
 - Two entities are associated
- ❑ Ternary relationship
 - Three entities are associated
 - Not necessarily equivalent to several 1:M relationships

Relationship Degree

- ❑ Crow's foot model does not allow implementation of M:N, additional entities are required
 - Copy of the same entity in a unary relationship (COURSE)
 - The CFR relationship in the Chen model is converted to an CFR entity in the Crow's foot model

Three Types of Relationships

FIGURE 4.16 THREE TYPES OF RELATIONSHIPS



The Implementation of a Ternary Relationship

FIGURE 4.17 THE IMPLEMENTATION OF A TERNARY RELATIONSHIP

Database name: Ch04_MedCo

Table name: CONTRIBUTOR

	CONTRIB_ID	CONTRIB_LNAME
►	C1	Brown
	C2	Iglesas
	C3	Smith

Table name: FUND

	FUND_ID	FUND_NAME	CONTRIB_ID	FUND_AMOUNT
►	F1	Heart	C1	\$50,000.00
	F1	Heart	C2	\$10,000.00
	F2	Cancer	C1	\$10,000.00
	F2	Cancer	C2	\$5,000.00
	F2	Cancer	C3	\$10,000.00

Table name: RECIPIENT

	REC_ID	REC_TYPE
►	R1	Rogers
	R2	Chen
	R3	Oshanski

Table name: CFR

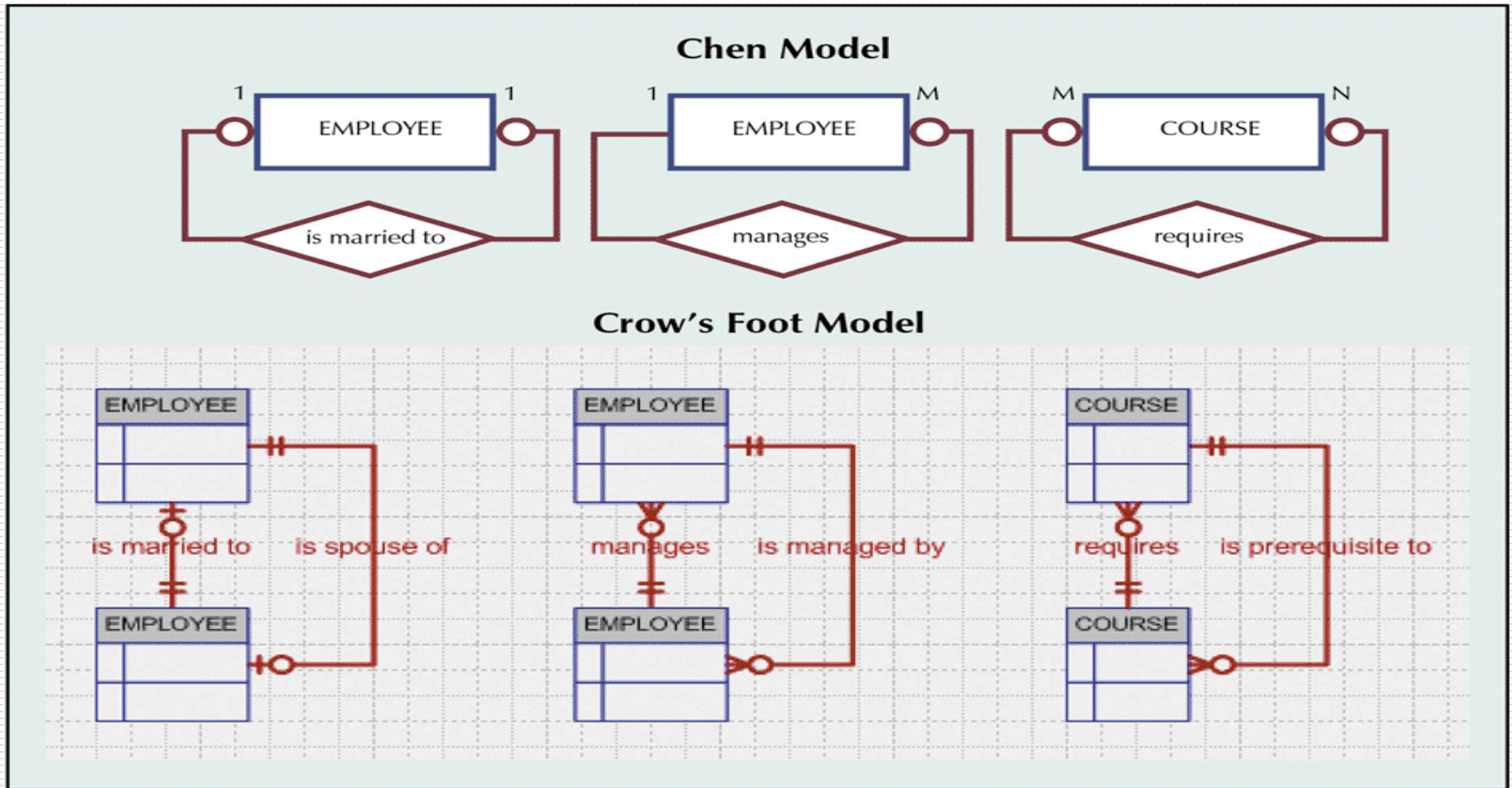
	FUND_ID	CON_ID	REC_ID	CFR_AMOUNT
►	F1	C1	R2	\$30,000.00
	F1	C1	R3	\$20,000.00
	F1	C2	R2	\$10,000.00
	F2	C1	R1	\$10,000.00
	F2	C2	R1	\$5,000.00

Recursive Relationships

- ❑ Relationship can exist between occurrences of the same entity set
- ❑ Naturally found within a unary relationship
 - 1:1 - EMPLOYEE is married to one and only one other EMPLOYEE
 - 1:M - An EMPLOYEE may manage many EMPLOYEEs and each EMPLOYEE is managed by one EMPLOYEE
 - M:N - A COURSE may be a prerequisite to many other COURSEs and each COURSE may have many other COURSEs as prerequisites

An ER Representation of Recursive Relationships

FIGURE 4.18 AN ER REPRESENTATION OF RECURSIVE RELATIONSHIPS



The 1:1 Recursive Relationship “EMPLOYEE is Married to EMPLOYEE”

FIGURE 4.19 THE 1:1 RECURSIVE RELATIONSHIP “EMPLOYEE IS MARRIED TO EMPLOYEE”

Table name: EMPLOYEE_V1

Database name: Ch04_PartCo

	EMP_NUM	EMP_LNAME	EMP_FNAME	EMP_SPOUSE
▶	345	Ramirez	James	347
	346	Jones	Anne	349
	347	Ramirez	Louise	345
	348	Delaney	Robert	
	349	Shapiro	Anton	346

Implementation of the 1:M “EMPLOYEE Manages EMPLOYEE” Recursive Relationship

FIGURE 4.23 IMPLEMENTATION OF THE 1:M “EMPLOYEE MANAGES EMPLOYEE” RECURSIVE RELATIONSHIP

Table name: EMPLOYEE_V2

Database name: Ch04_PartCo

	EMP_CODE	EMP_LNAME	EMP_MANAGER
▶	101	Waddell	102
	102	Orincona	
	103	Jones	102
	104	Reballoh	102
	105	Robertson	102
	106	Deltona	102

Implementation of the M:N Recursive “COURSE Requires COURSE” Relationship

COURSE

CRS_CODE	DEPT_CODE	CRS_DESCRIPTION	CRS_CREDIT
ACCT-211	ACCT	Accounting I	3
ACCT-212	ACCT	Accounting II	3
CIS-220	CIS	Intro. to Microcomputing	3
CIS-420	CIS	Database Design and Implementation	4
MATH-243	MATH	Mathematics for Managers	3
QM-261	CIS	Intro. to Statistics	3
QM-362	CIS	Statistical Applications	4

PREREQ

CRS_CODE	PRE_TAKE
CIS-420	CIS-220
QM-261	MATH-243
QM-362	MATH-243
QM-362	QM-261

❑ MATH-243 is a prerequisite to QM-261 and QM-362

❑ MATH-243 and QM-261 are prerequisites to QM-362

Implementation of the M:N Recursive “PART Contains PART” Relationship

FIGURE 4.21 IMPLEMENTATION OF THE M:N RECURSIVE “PART CONTAINS PART” RELATIONSHIP

Table name: COMPONENT

Database name: Ch04_PartCo

	COMP_CODE	PART_CODE	COMP_PARTS_NEEDED
▶	C-130	AA21-6	4
	C-130	AB-121	2
	C-130	E129	1
	C-131A2	E129	1
	C-130	X10	4
	C-131A2	X10	1
	C-130	X34AV	2
	C-131A2	X34AV	2

Table name: PART

	PART_CODE	PART_DESCRIPTION	PART_IN_STOCK
▶	AA21-6	2.5 cm. washer, 1.0 mm. rim	432
	AB-121	Cotter pin, copper	1,034
	C-130	Rotor assembly	36
	E129	2.5 cm. steel shank	128
	X10	10.25 cm. rotor blade	345
	X34AV	2.5 cm. hex nut	879

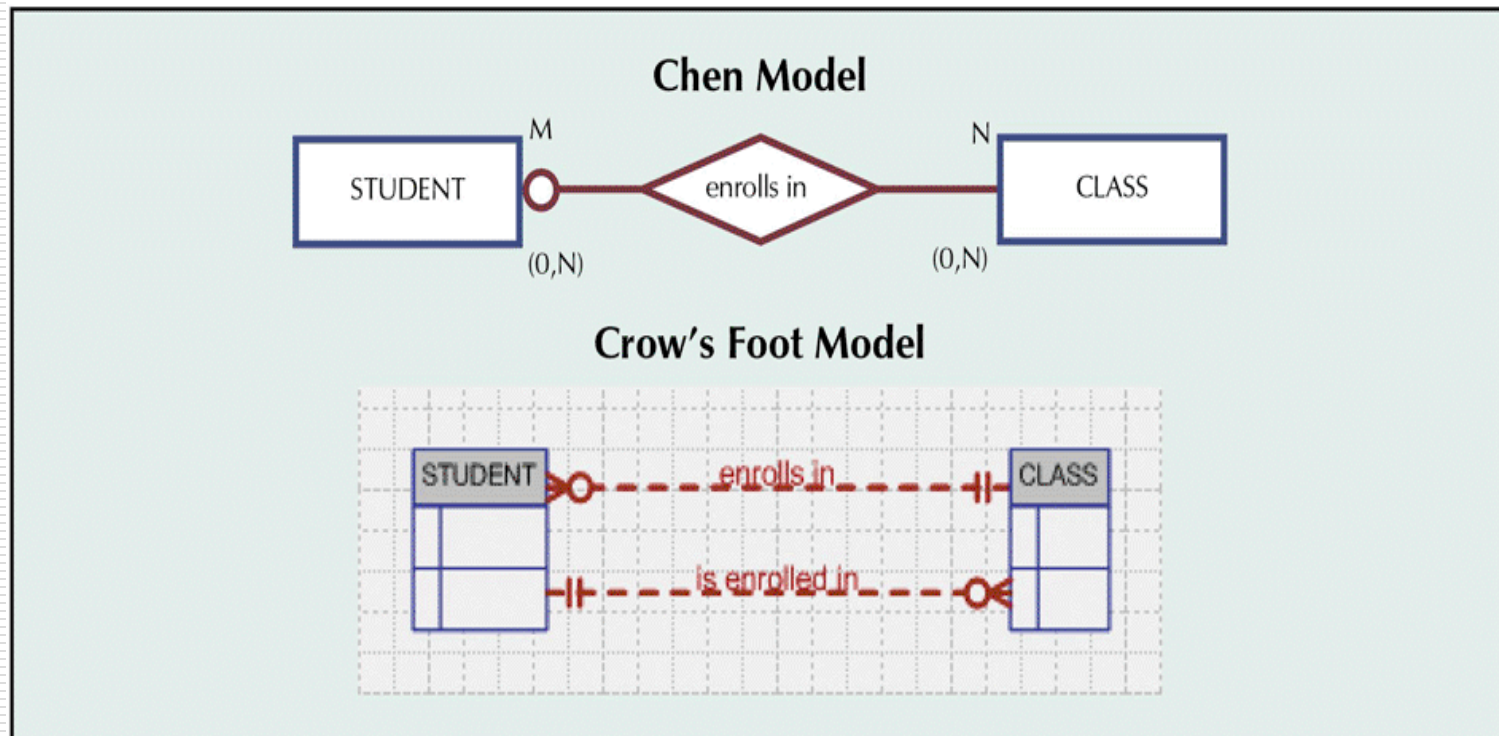
Composite Entities

- ❑ Also known as *bridge entities*
- ❑ Composed of the primary keys of each of the entities to be connected
- ❑ May also contain additional attributes that play no role in the connective process

The M:N Relationship Between STUDENT and CLASS

- ❑ A class may exist even though it contains no students, thus the *optional* symbol appears on the STUDENT side of the M:N relationship

FIGURE 4.25 THE M:N RELATIONSHIP BETWEEN STUDENT AND CLASS



Converting the M:N Relationship into Two 1:M Relationships

FIGURE 4.24 CONVERTING THE M:N RELATIONSHIP INTO TWO 1:M RELATIONSHIPS

Table name: STUDENT

Database name: Ch04_CollegeTry

	STU_NUM	STU_LNAME
▶ +	321452	Bowser
+	324257	Smithson

Table name: ENROLL

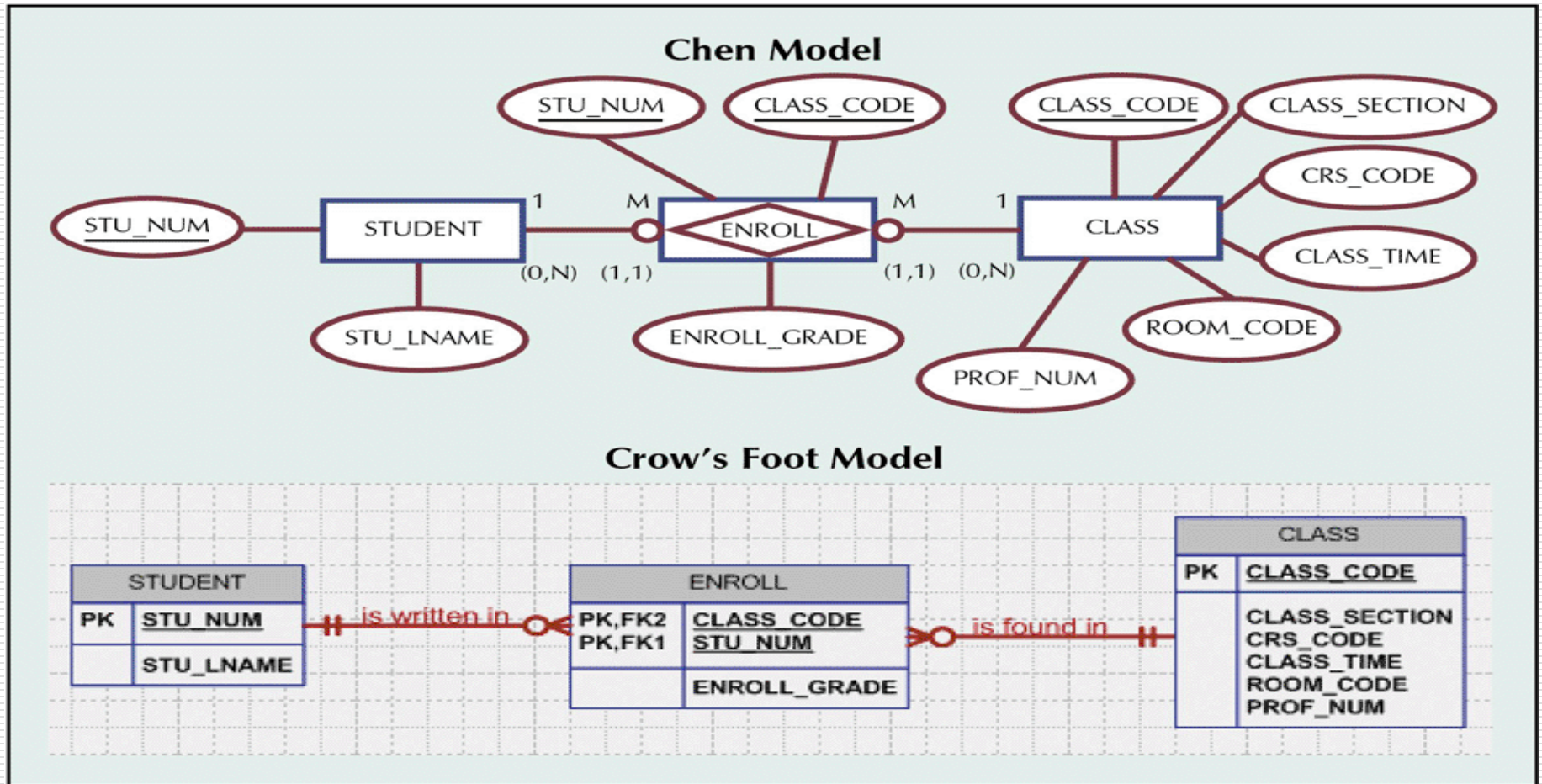
	CLASS_CODE	STU_NUM	ENROLL_GRADE
▶	10014	321452	C
	10014	324257	B
	10018	321452	A
	10018	324257	B
	10021	321452	C
	10021	324257	C

Table name: CLASS

	CLASS_CODE	CRS_CODE	CLASS_SECTION	CLASS_TIME	CLASS_ROOM	PROF_NUM
▶ +	10014	ACCT-211	3	TTh 2:30-3:45 p.m.	BUS252	342
+	10018	CIS-220	2	MWTF 9:00-9:50 a.m.	KLR211	114
+	10021	QM-261	1	MWTF 8:00-8:50 a.m.	KLR200	114

A Composite Entity in an ERD

FIGURE 4.26 A COMPOSITE ENTITY IN AN ERD



Nulls Created by Unique Attributes

- Company has many types of employees, they have attributes in common and attributes unique to each type
- One table for all employees could have many nulls and/or dummy entries for the unique attributes not used by other types of employees

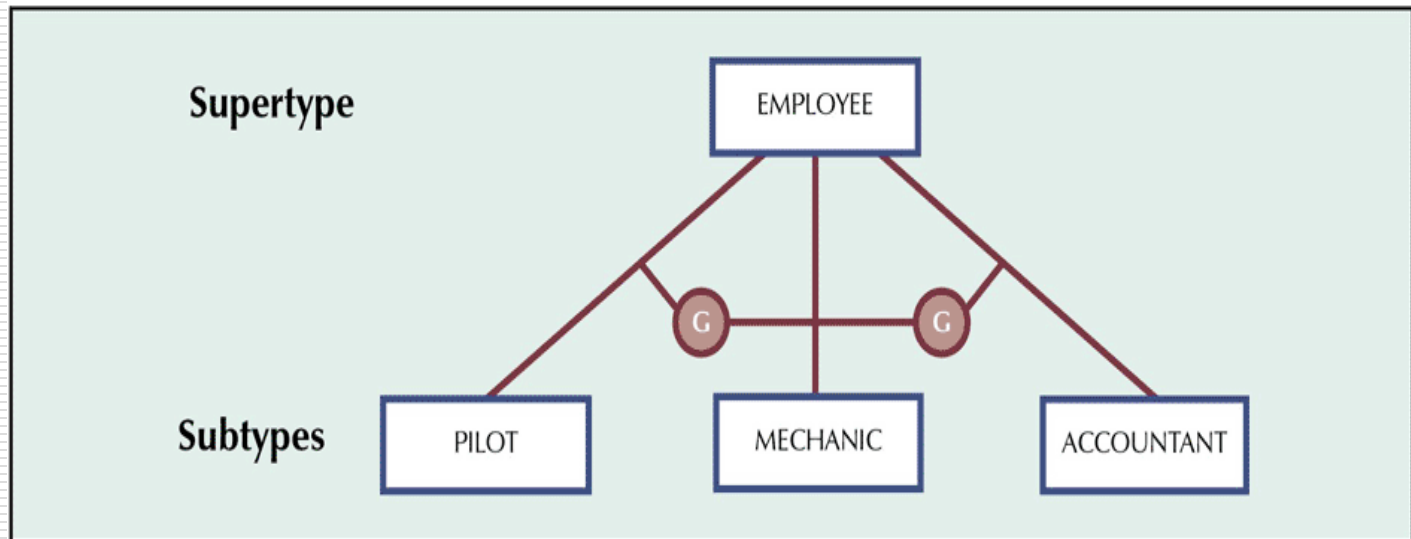
FIGURE 4.27 NULLS CREATED BY UNIQUE ATTRIBUTES

Table name: EMPLOYEE_V1		Database name: Ch04_AirCo				
	EMP_NUM	EMP_LNAME	EMP_LICENSE	EMP_RATINGS	EMP_MED_TYPE	EMP_HIRE_DATE
▶	100	Kolmycz				15-Mar-88
	101	Lewis	ATP	SEL/MEL/Instr/CFII	1	25-Apr-89
	102	Vandam				20-Dec-93
	103	Jones				28-Aug-03
	104	Lange	ATP	SEL/MEL/Instr	1	20-Oct-97
	105	Williams	COM	SEL/MEL/Instr/CFI	2	08-Nov-97
	106	Duzak	COM	SEL/MEL/Instr	2	05-Jan-04
	107	Diante				02-Jul-97
	108	Wiesenbach				18-Nov-95
	109	Travis	COM	SEL/MEL/SES/Instr/CFII	1	14-Apr-01
	110	Genkazi				01-Dec-03

A Generalization Hierarchy

- Depicts a relationship between a higher-level supertype entity and a lower-level subtype entity
 - Supertype entity contains shared attributes
 - Subtype entity contains unique attributes

FIGURE 4.28 A GENERALIZATION HIERARCHY



Disjoint Subtypes

- ❑ Also known as non-overlapping subtypes
 - Subtypes that contain a subset of the supertype entity set
 - Each entity instance (row) of the supertype can appear in only one of the disjoint subtypes
 - Denoted by the symbol  on the model
 - ❑ Employee can be a pilot but not, at the same time, an accountant
- ❑ Supertype and its subtype(s) maintain a 1:1 relationship

The EMPLOYEE/PILOT Supertype/Subtype Relationship

FIGURE 4.29 THE EMPLOYEE/PILOT SUPERTYPE/SUBTYPE RELATIONSHIP

Table name: EMPLOYEE (the supertype)

Database name: Ch04_AirCo

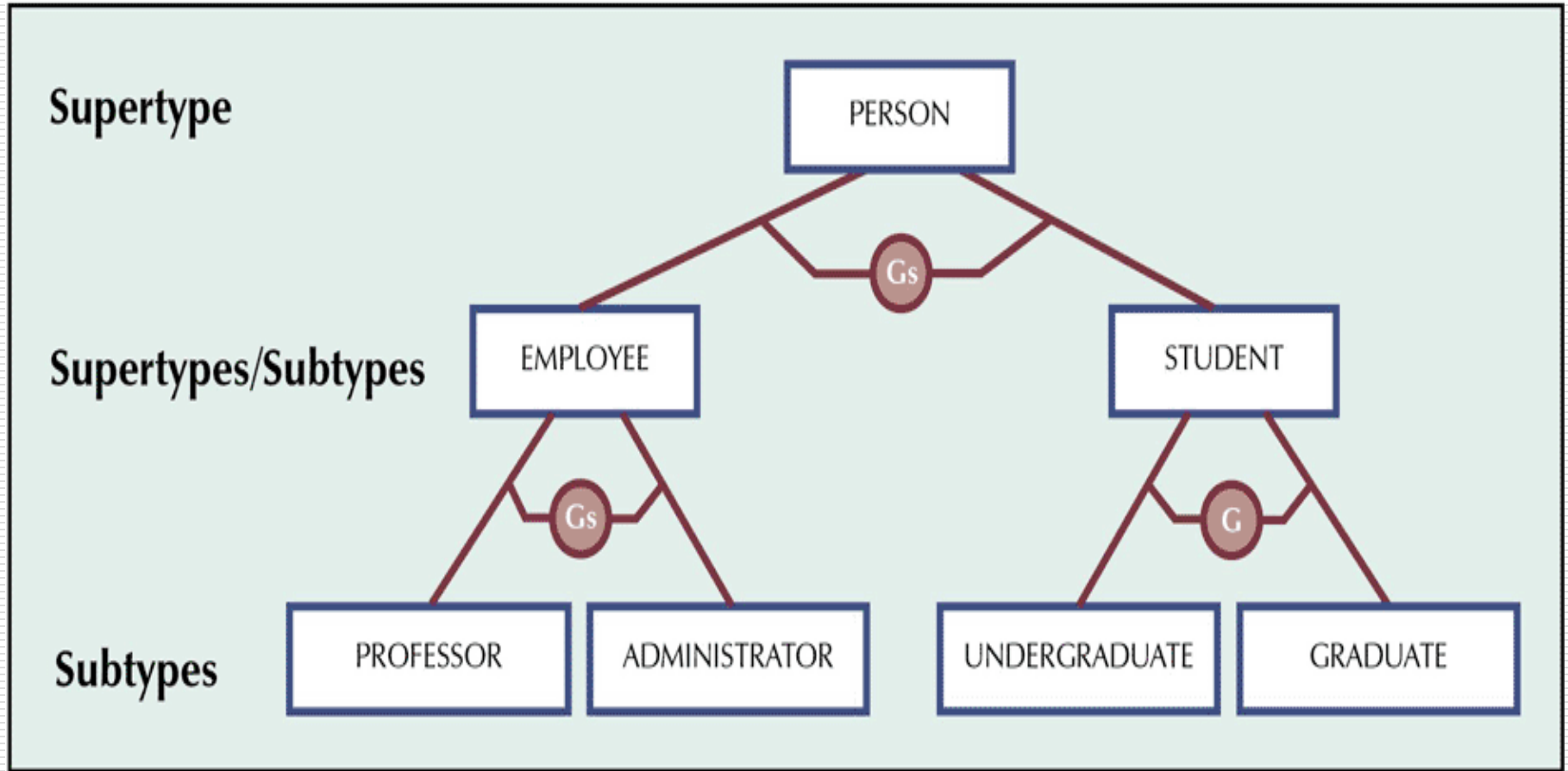
		EMP_NUM	EMP_LNAME	EMP_HIRE_DATE
▶	+	100	Kolmycz	15-Mar-88
	+	101	Lewis	25-Apr-89
	+	102	Vandam	20-Dec-93
	+	103	Jones	28-Aug-03
	+	104	Lange	20-Oct-97
	+	105	Williams	08-Nov-97
	+	106	Duzak	05-Jan-04
	+	107	Diante	02-Jul-97
	+	108	Wiesenbach	18-Nov-95
	+	109	Travis	14-Apr-01
	+	110	Genkazi	01-Dec-03

Table name: PILOT (the subtype)

		EMP_NUM	PIL_LICENSE	PIL_RATINGS	PIL_MED_TYPE
▶	+	101	ATP	SEL/MEL/Instr/CFII	1
	+	104	ATP	SEL/MEL/Instr	1
	+	105	COM	SEL/MEL/Instr/CFI	2
	+	106	COM	SEL/MEL/Instr	2
	+	109	COM	SEL/MEL/SES/Instr/CFII	1

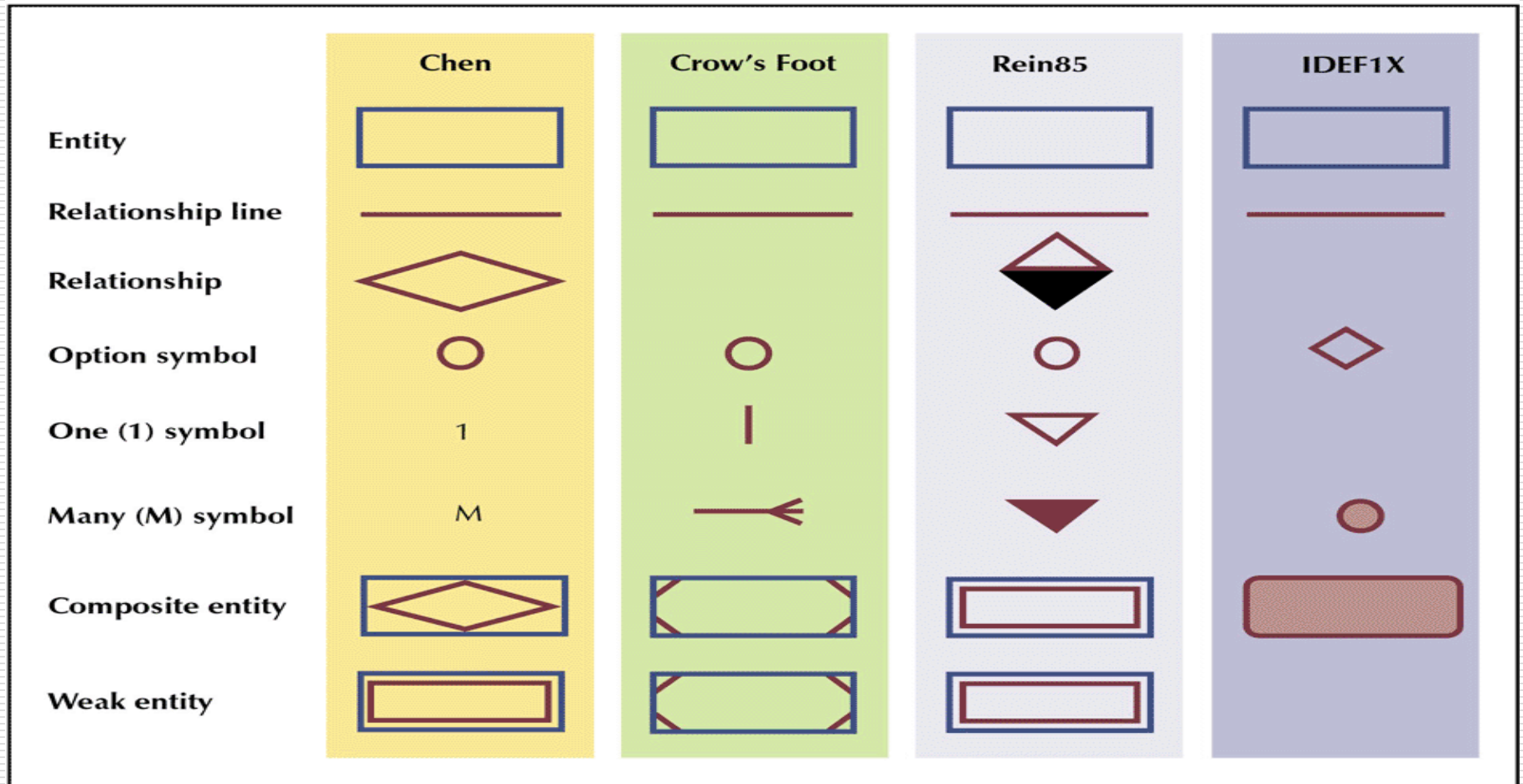
A Generalization Hierarchy with Overlapping Subtypes

FIGURE 4.30 A GENERALIZATION HIERARCHY WITH OVERLAPPING SUBTYPES



A Comparison of ER Modeling Symbols

FIGURE 4.31 A COMPARISON OF ER MODELING SYMBOLS



A Comparison of ER Modeling Symbols

- ❑ **Chen model** moved conceptual modeling into the practical database design arena by establishing basic building blocks: entities and relationships
 - Dominant player in the CASE tool market during the 1980s and early 1990s
- ❑ **Crow's Foot model** combines connectivity and cardinality information in a single symbol set. Popularized by the Knowledgeware modeling tool
 - Cardinality is limited to 0,1 or N

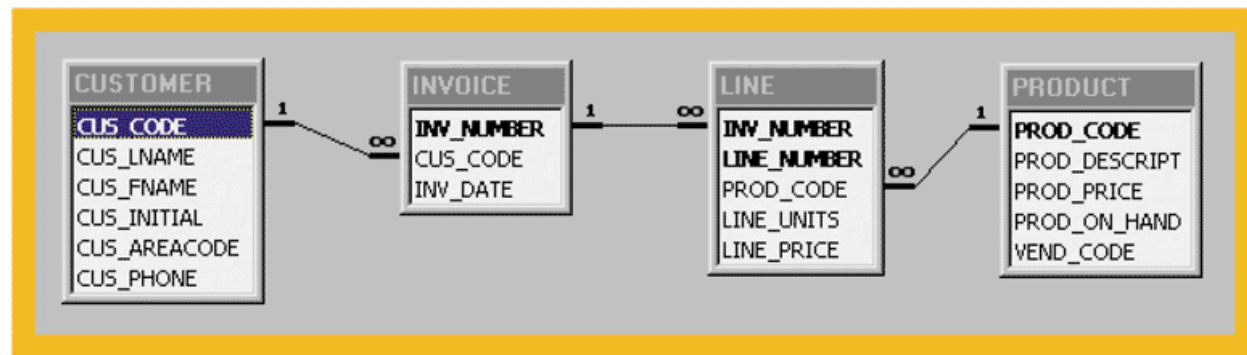
A Comparison of ER Modeling Symbols

- ❑ **Rein85 model** based on the same modeling conventions as the Crow's Foot model, it's symbols are different.
 - It does not recognize cardinalities explicitly, relying on connectivities to lead to logical cardinality conclusions
- ❑ **IDEF1X** is a derivative of the integrated computer-aided manufacturing (ICAM) studies of the late 1970s.
 - Became the source of graphical methods for defining the functions, data structures and dynamics of manufacturing businesses.
 - The integration of these methods became known as IDEF(ICAM Definition). Hughes Aircraft developed the original version named IDEF1. The extended version, known as IDEF1X, became the USAF standard

The Chen Representation of the Invoicing Problem

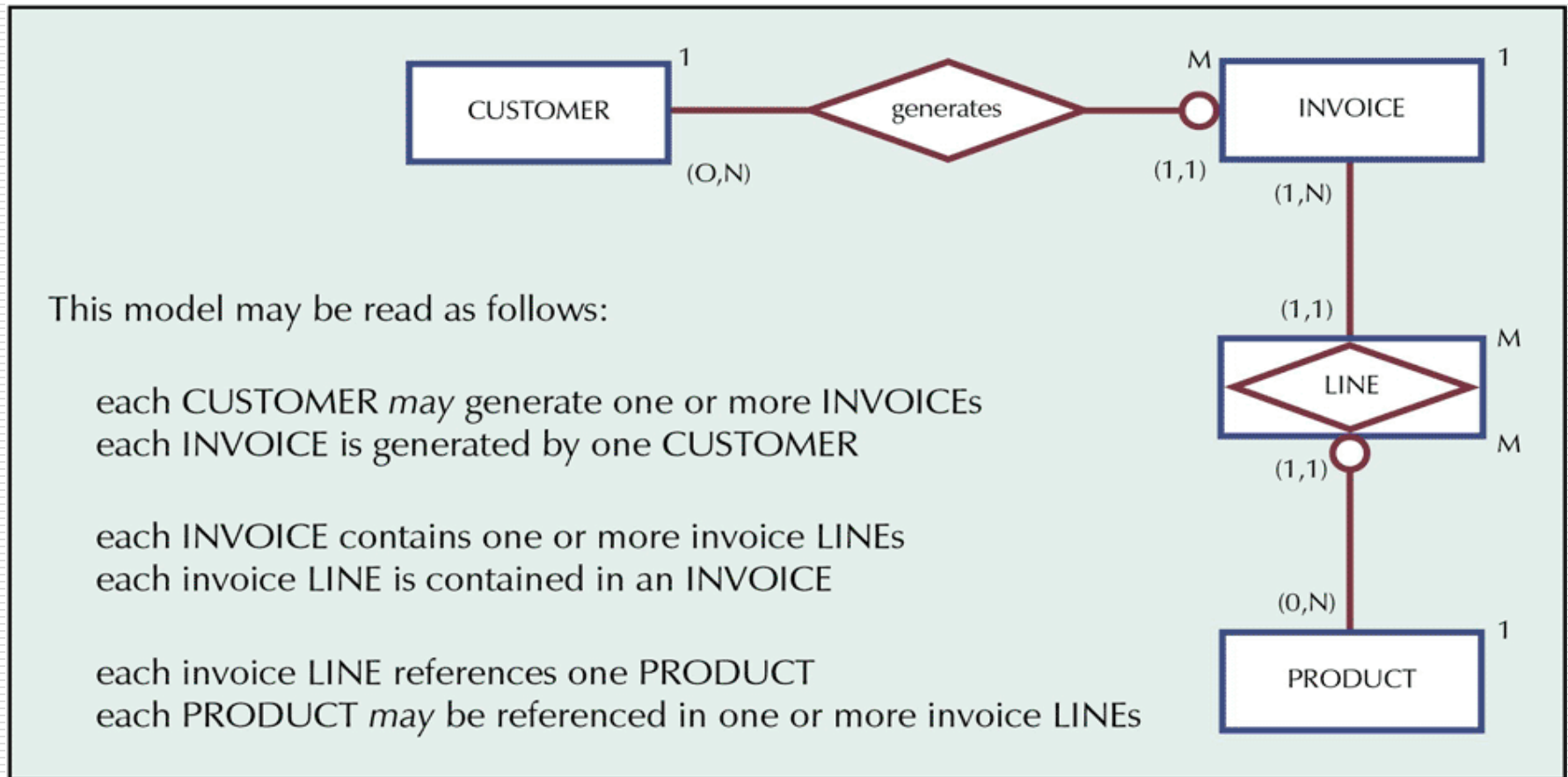
- A customer may not have made a purchase so INVOICE is optional to CUSTOMER
- Some products kept in inventory are never sold and may never show up in an invoice. INVOICE is optional to PRODUCT (M:N)
- Because LINE is used to decompose the M:N relationship into two 1:M relationships, LINE becomes optional to PRODUCT

FIGURE 3.31 THE RELATIONAL SCHEMA FOR THE INVOICING SYSTEM



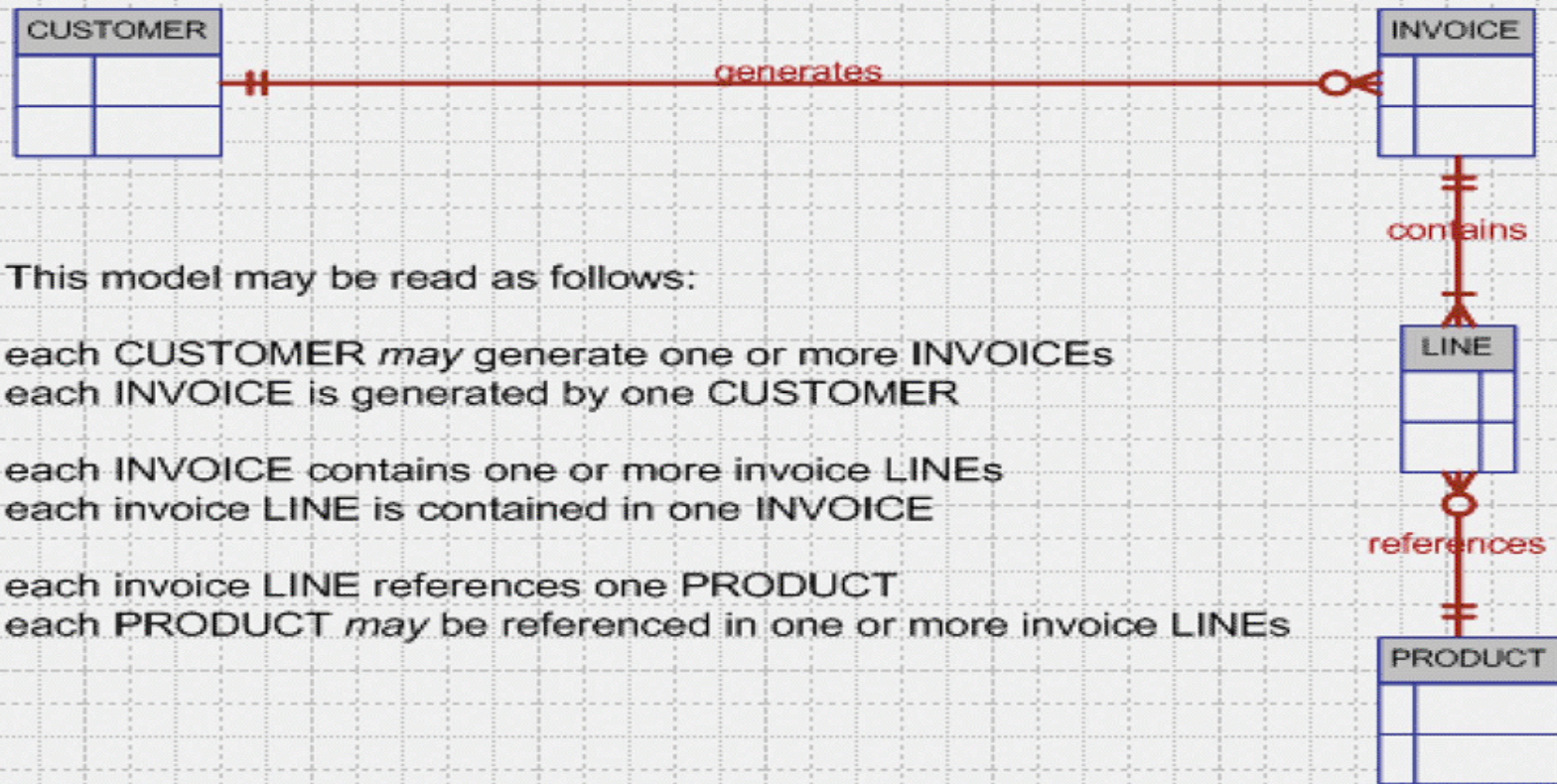
The Chen Representation of the Invoicing Problem

FIGURE 4.32 THE CHEN REPRESENTATION OF THE INVOICING PROBLEM



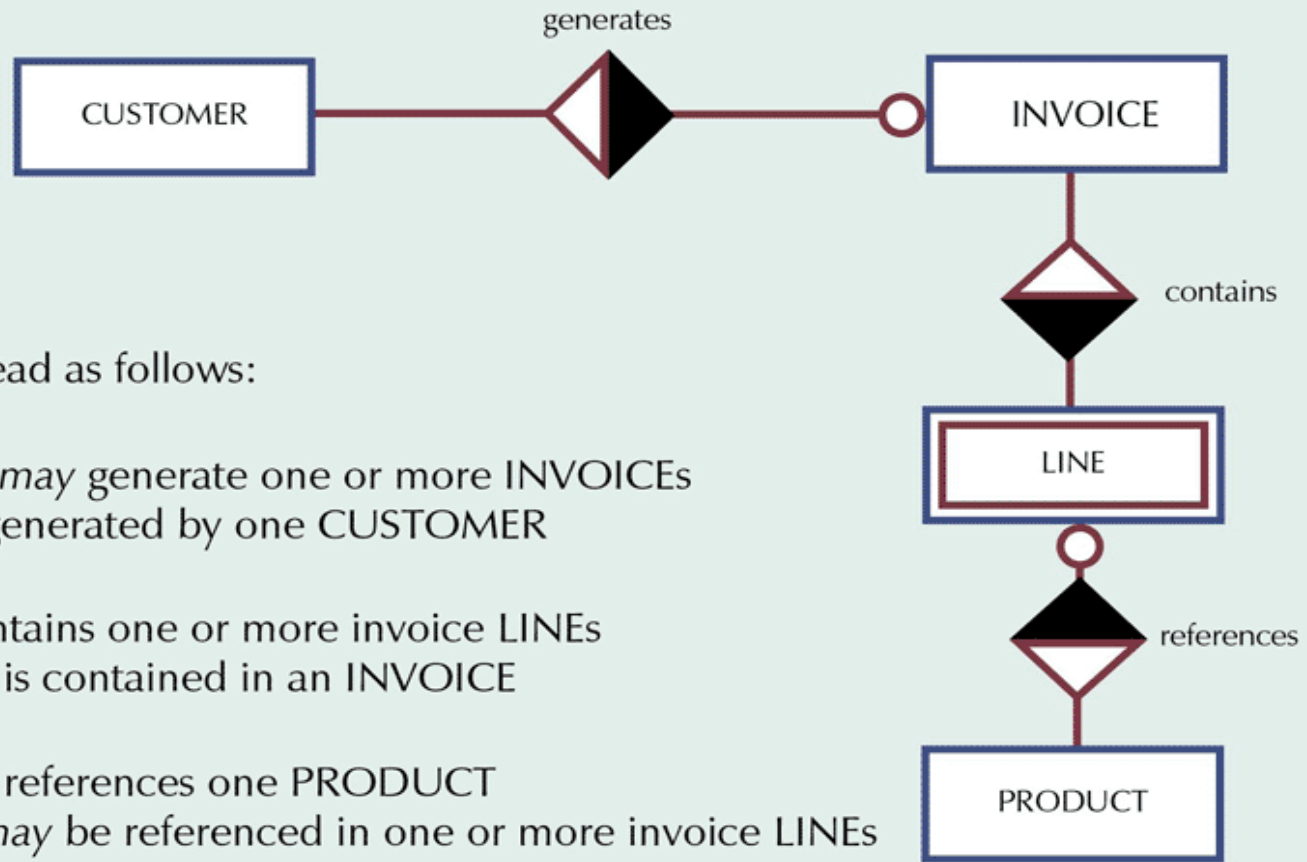
The Crow's Foot Representation of the Invoicing Problem

FIGURE 4.33 THE CROW'S FOOT REPRESENTATION OF THE INVOICING PROBLEM



The Rein85 Representation of the Invoicing Problem

FIGURE 4.34 THE REIN85 REPRESENTATION OF THE INVOICING PROBLEM



This model may be read as follows:

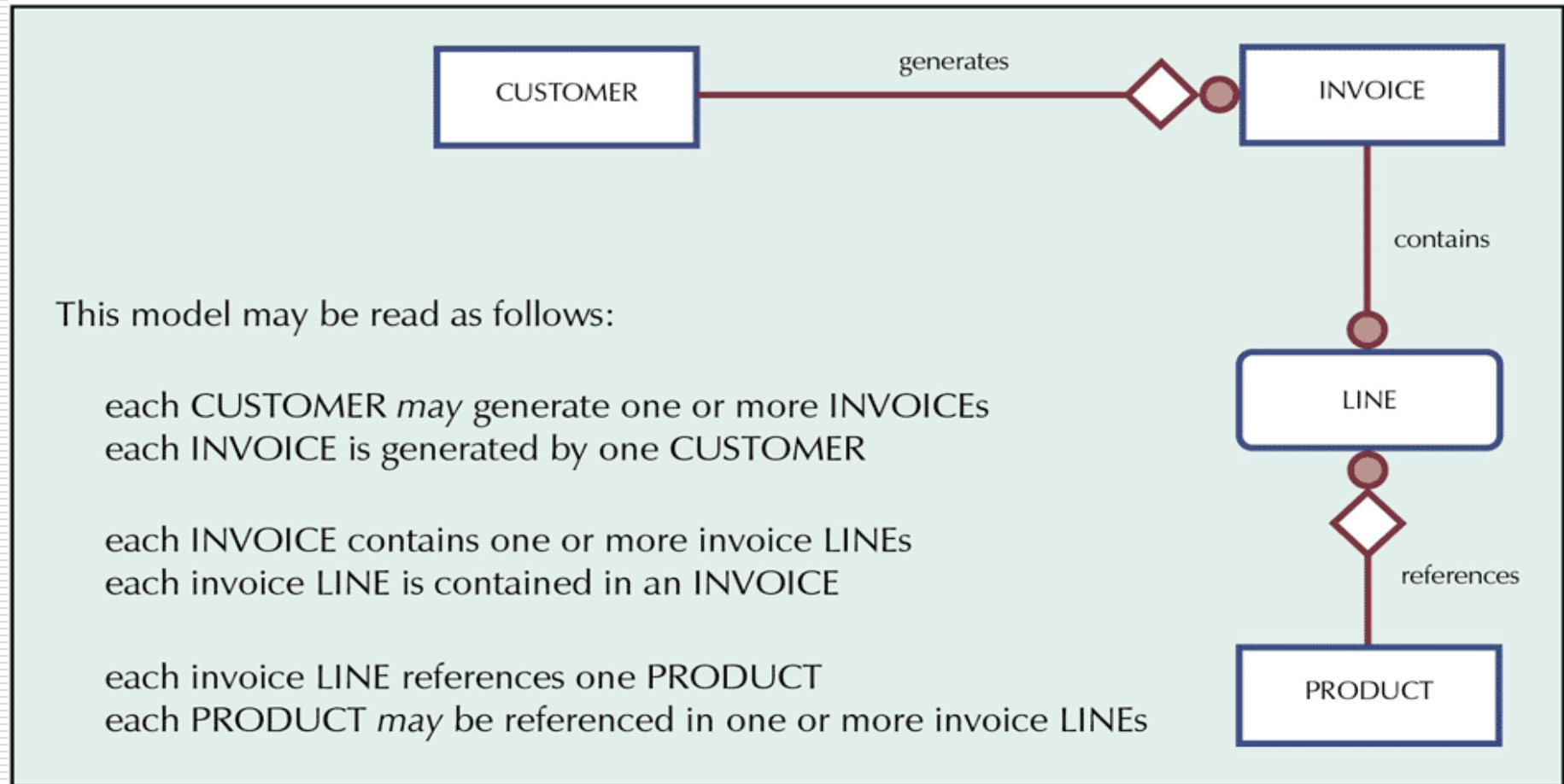
each CUSTOMER *may* generate one or more INVOICES
each INVOICE is generated by one CUSTOMER

each INVOICE contains one or more invoice LINES
each invoice LINE is contained in an INVOICE

each invoice LINE references one PRODUCT
each PRODUCT *may* be referenced in one or more invoice LINES

The IDEF1X Representation of the Invoicing Problem

FIGURE 4.35 THE IDEF1X REPRESENTATION OF THE INVOICING PROBLEM

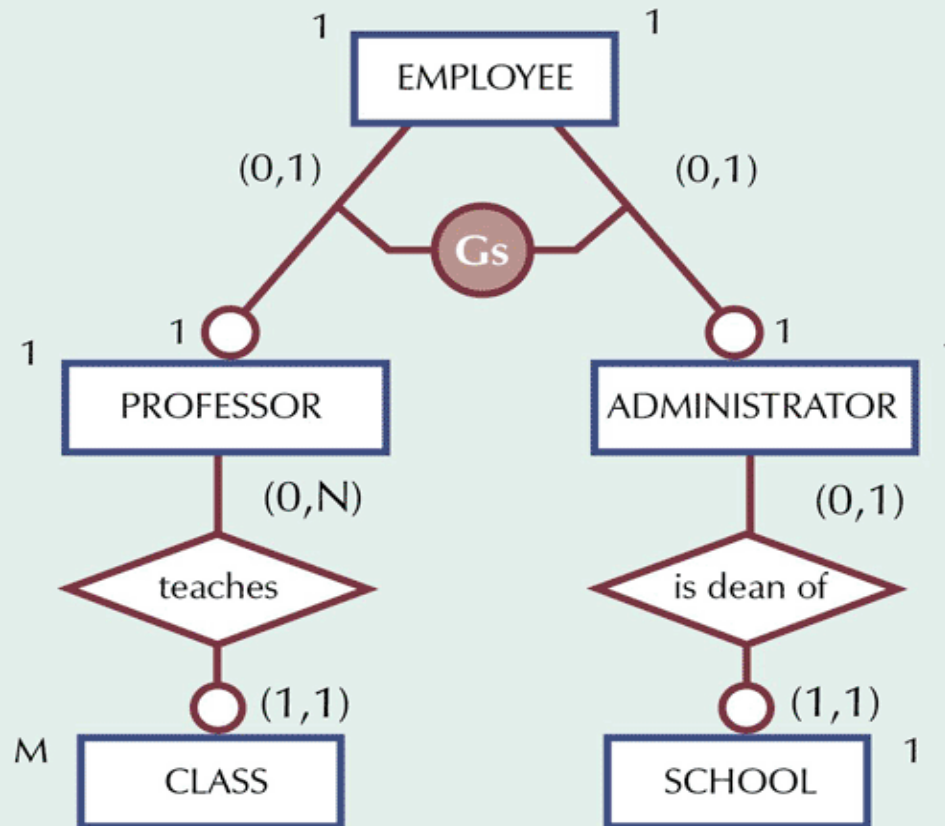


Developing an ER Diagram

- ❑ Database design is an iterative rather than a linear or sequential process
- ❑ Information gathered from interviews but also by examining business forms and reports used on a daily basis
- ❑ Iterative process
 - Based on repetition of processes and procedures

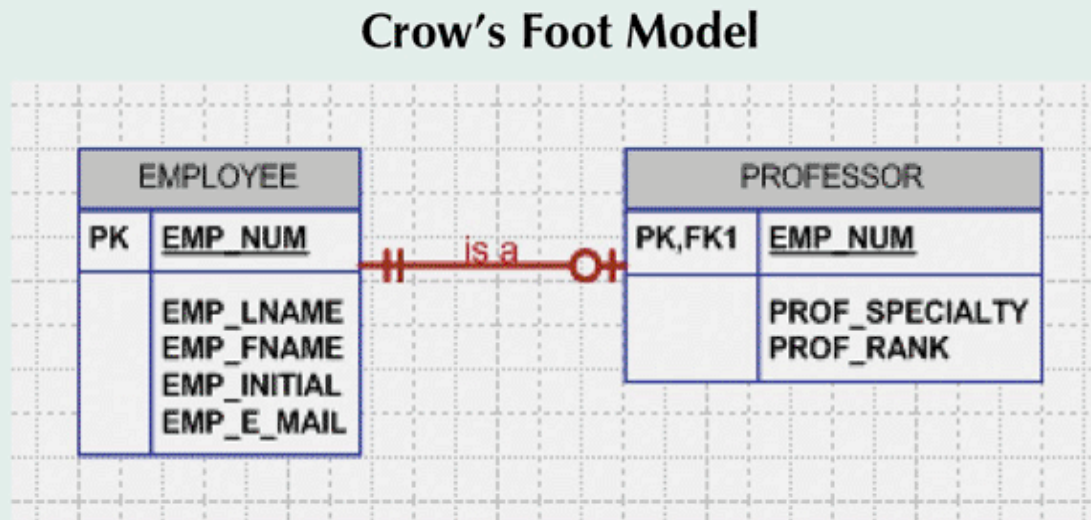
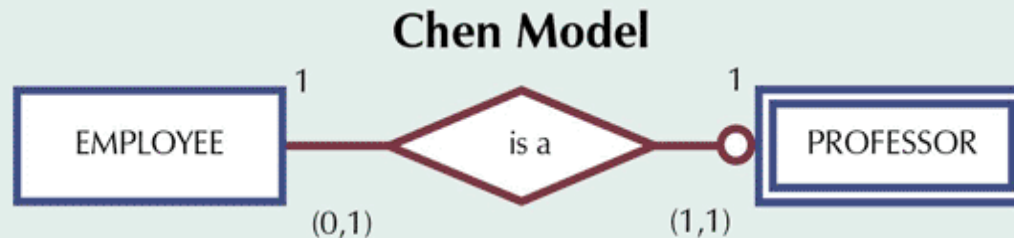
A Supertype/Subtype Relationship

FIGURE 4.36 A SUPERTYPE/SUBTYPE RELATIONSHIP



A Supertype/Subtype Relationship in an ERD

FIGURE 4.37 A SUPERTYPE/SUBTYPE RELATIONSHIP IN AN ERD



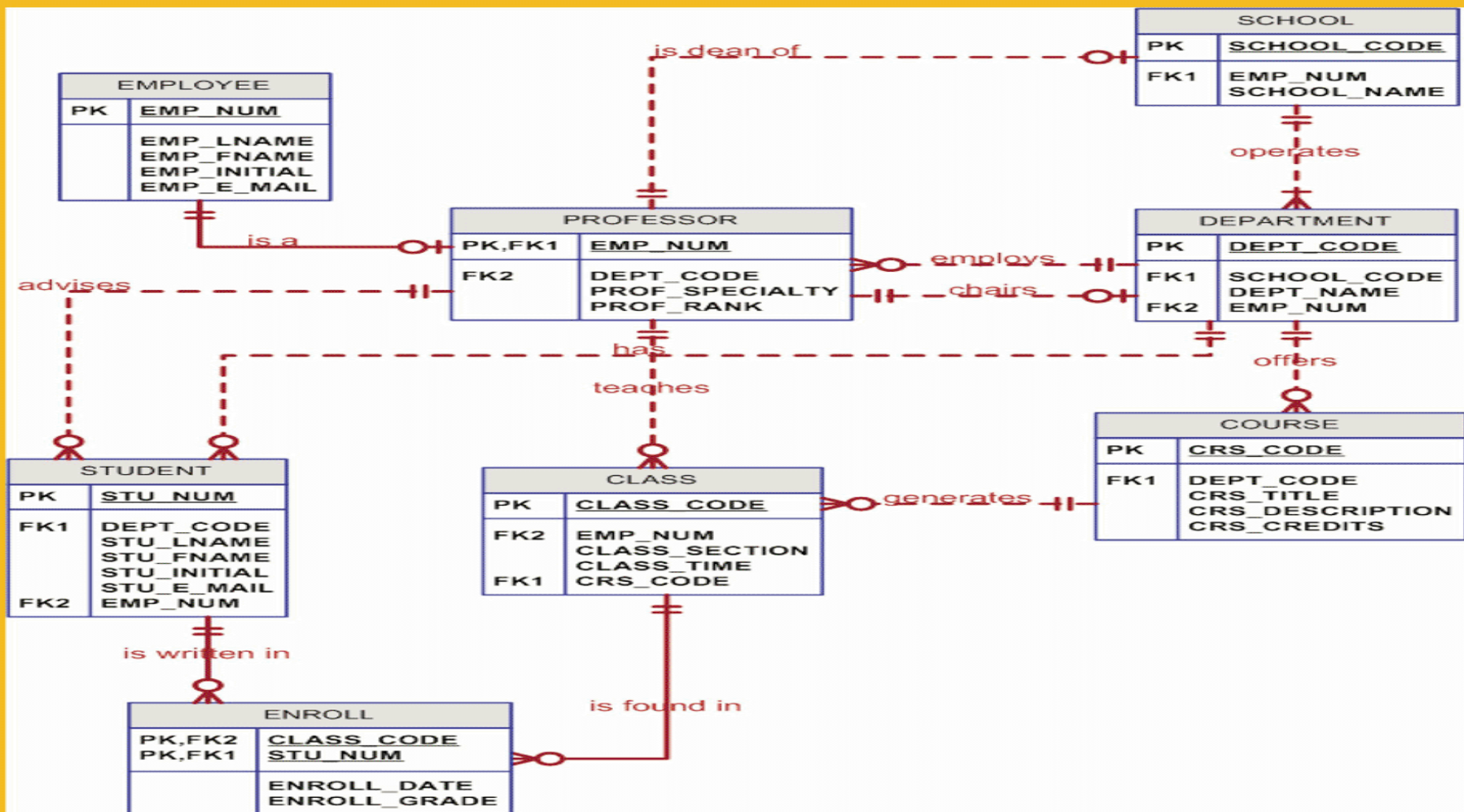
Components of the ER Model

TABLE 4.2 COMPONENTS OF THE ER MODEL

ENTITY	RELATIONSHIP	CONNECTIVITY	ENTITY
SCHOOL	operates	1:M	DEPARTMENT
DEPARTMENT	has	1:M	STUDENT
DEPARTMENT	employs	1:M	PROFESSOR
DEPARTMENT	offers	1:M	COURSE
COURSE	generates	1:M	CLASS
PROFESSOR	is an	1:1	EMPLOYEE
PROFESSOR	is dean of	1:1	SCHOOL
PROFESSOR	chairs	1:1	DEPARTMENT
PROFESSOR	teaches	1:M	CLASS
PROFESSOR	advises	1:M	STUDENT
STUDENT	enrolls in	1:M	CLASS
BUILDING	contains	1:M	ROOM
ROOM	is used for	1:M	CLASS

The Completed Tiny College ERD

FIGURE 4.47 THE COMPLETED TINY COLLEGE ERD



The Challenge of Database Design: Conflicting Goals

- ❑ Database design must conform to design standards
- ❑ High processing speeds are often a top priority in database design
- ❑ Quest for timely information might be the focus of database design
 - Sacrificing some of the “clean” design structures and/or high transaction speed may necessary to ensure maximum information generation
 - ❑ Instead of generating taxes, subtotals, totals, etc. each time a report is printed, compute and store these derived values
- ❑ Other issues to consider– security, data integrity, performance, shared access

Various Implementations of a 1:1 Recursive Relationship

FIGURE 4.48 VARIOUS IMPLEMENTATIONS OF A 1:1 RECURSIVE RELATIONSHIP

Table name: EMPLOYEE_V1

	EMP_NUM	EMP_LNAME	EMP_FNAME	EMP_SPOUSE
▶	345	Ramirez	James	347
	346	Jones	Anne	349
	347	Ramirez	Louise	345
	348	Delaney	Robert	
	349	Shapiro	Anton	346

Database name: Ch04_PartCo

First implementation

Table name: EMPLOYEE

	EMP_NUM	EMP_LNAME	EMP_FNAME
▶	345	Ramirez	James
	346	Jones	Anne
	347	Ramirez	Louise
	348	Delaney	Robert
	349	Shapiro	Anton

Table name: MARRIED_V1

	EMP_NUM	EMP_SPOUSE
▶	345	347
	346	349
	347	345
	349	346

Second implementation

Table name: MARRIAGE

	MAR_NUM	MAR_DATE
▶	1	04-Mar-03
	2	02-Feb-99

Table name: MARPART

	MAR_NUM	EMP_NUM
▶	1	345
	1	347
	2	346
	2	349

Table name: EMPLOYEE

	EMP_NUM	EMP_LNAME	EMP_FNAME
▶	345	Ramirez	James
	346	Jones	Anne
	347	Ramirez	Louise
	348	Delaney	Robert
	349	Shapiro	Anton

The Relational Schema for the Third Implementation



Third implementation

Various Implementations of a 1:1 Recursive Relationship

- ❑ EMPLOYEE_V1 may cause anomalies
 - If employees divorce, two records must be updated
 - Those not married to other employees have a null entry
 - Uses synonyms to refer to an employee – EMP_NUM and EMP_SPOUSE
- ❑ MARRIED_V1
 - Eliminates nulls but duplicate values are still possible (345,347) and (347,345) – each is unique
 - Uses synonyms to refer to an employee – EMP_NUM and EMP_SPOUSE
- ❑ Three table solution
 - Add MARRIAGE and MARPART tables in a 1:M relationship
 - ❑ Make EMP_NUM unique in MARPART so that an employee does not appear twice as married